VITCC CENTRAL LIBRARY MANAGEMENT SYSTEM

MULTI-ROLE INTERFACE AUDIT & ROLE-BASED ACCESS CONTROL (RBAC) VERIFICATION REPORT

Automated high-fidelity capture of all system sections across Administrator, Librarian, Faculty/Staff, Student, and Public portals for autonomous LLM audit and compliance review.

Audit Date: October 18, 2026	**Target Environment:** Live Dev Server (http://127.0.0.1:5000)	**Display Viewport:** 1440 x 960 (Desktop HD)
Database Engines: Oracle Database 23ai + Neo4j Graph DB	**Total Views Audited:** 44 Unique Verified Screens	**Security Model:** Session RBAC + CSRF Protection

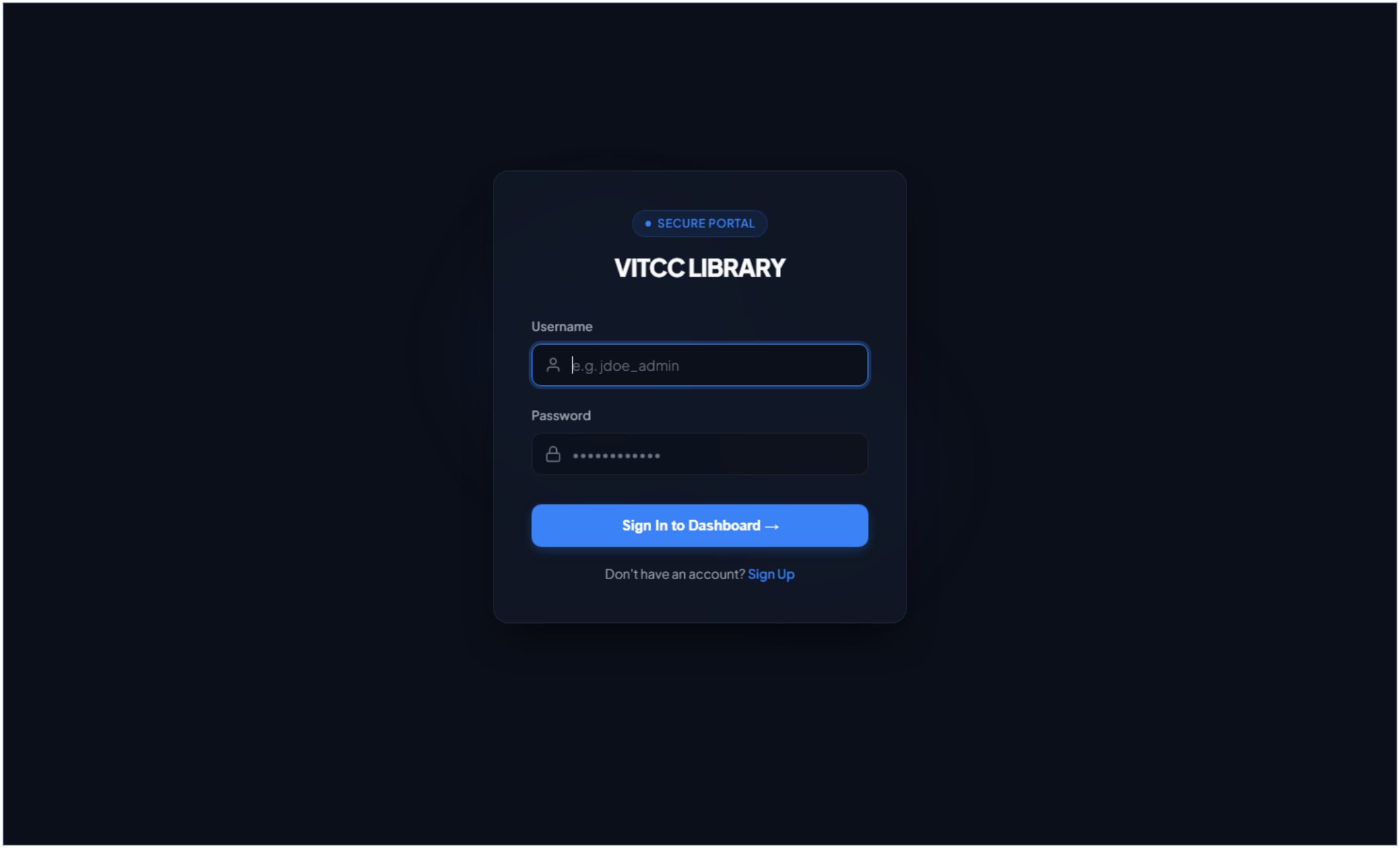
Application Section / Feature	Route URL	Public	Admin	Librarian	Staff	Student	Access Scope / Security Rule
Login & Authentication	/login	✓	✓	✓	✓	✓	Public credential verification & session issuing
Patron Registration	/register	✓	✓	✓	✓	✓	Self-service patron sign up (Student/Staff)
Dashboard & Overview	/dashboard	—	**✓ Full**	**✓ Desk**	**✓ Faculty**	**✓ Personal**	Role-tailored KPI cards, borrowing queues & charts
Catalog & Book Search	/books	—	✓	✓	✓	✓	Global catalog search with category & author filtering
Category & Curriculum Graph	/category/<id>	—	✓	✓	✓	✓	Prerequisite & unlocks badges, co-borrowing pills
Issue Book Loan	/issue_loan	—	**✓ Staffed**	**✓ Staffed**	**Blocked**	**Blocked**	Direct copy allocation desk (Admin & Librarian only)
Loan Requests & Collection	/loan_requests	—	**✓ Staffed**	**✓ Staffed**	**Blocked**	**Blocked**	Patron pickup approvals & collection desk handover
Manage Users & Governance	/manage_users	—	**✓ Admin**	**Blocked**	**Blocked**	**Blocked**	RBAC privileges, suspension & password resets (Admin)
Learning Roadmaps	/learning_paths	—	✓	✓	✓	✓	Curriculum tracks & multi-level study hierarchies
AI Recommendations	/recommendations	—	**✓ Trends**	**✓ Trends**	**✓ Faculty**	**✓ Peer**	Collaborative filtering, peer study partners & trends
My Fines & Settlements	/my_fines	—	—	—	**✓ Faculty**	**✓ Student**	Personal overdue fines & instant payment gateway
Knowledge Graph Explorer	/graph_visualizer	—	**✓ Global**	**✓ System**	**✓ Academic**	**✓ Learning**	Interactive Cytoscape / Vis.js network visualizer
Book Bridge (Shortest Path)	/book_bridge	—	**✓ Path**	**✓ Path**	**✓ Path**	**✓ Path**	Neo4j shortest path traversal between book nodes
Centrality Analytics Lab	/library_analytics	—	**✓ Lab**	**✓ Lab**	**✓ Lab**	**Blocked**	Author Degree, Betweenness & Closeness centrality
Change Password	/change_password	—	✓	✓	✓	✓	Self-service credential updates (8–64 character policy)

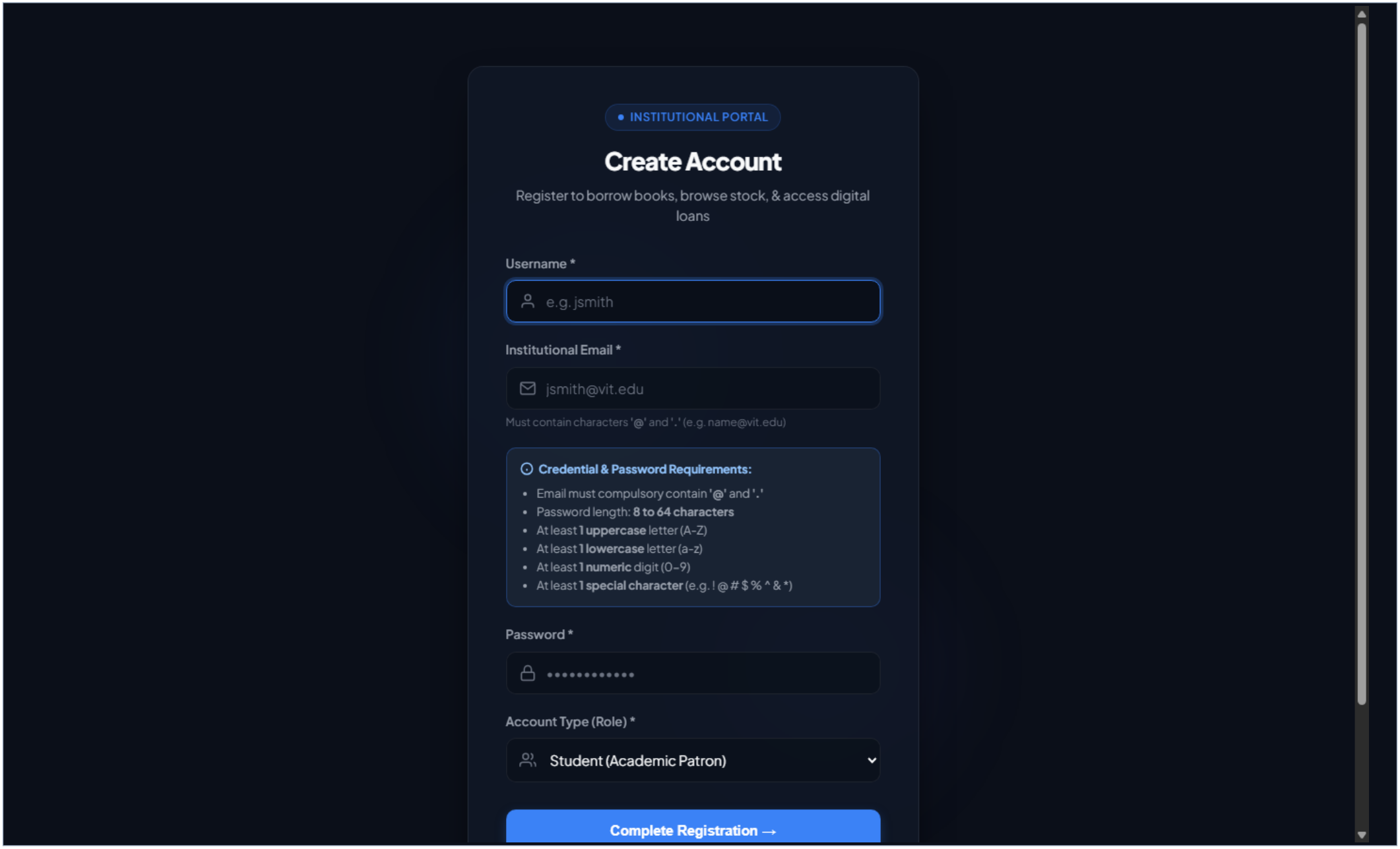
Login Authentication Portal

● PUBLIC ACCESS

Route: http://127.0.0.1:5000/login | **ID:** 01_public_login

VITCC Library UI Audit



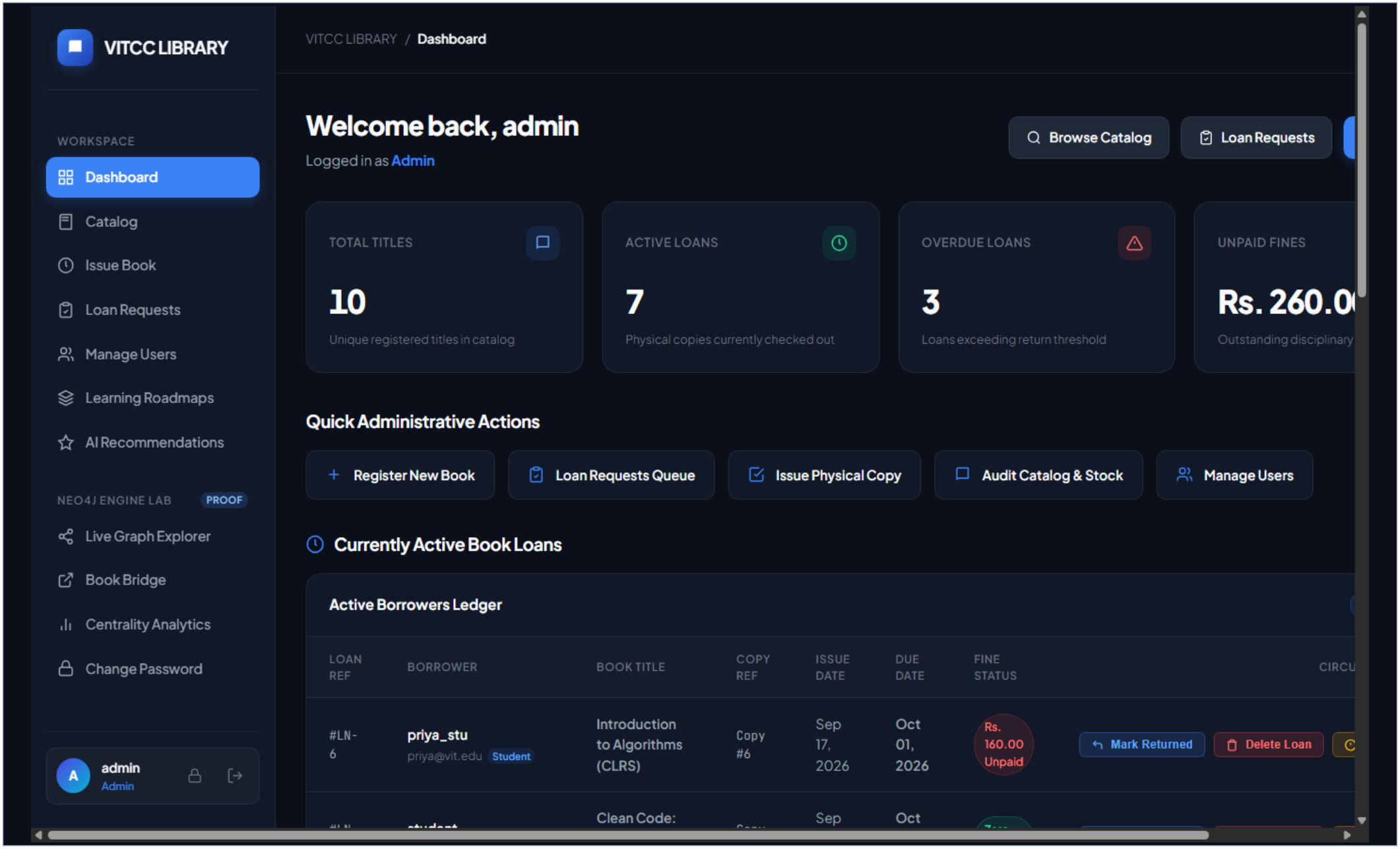

INSTITUTIONAL PORTAL
Create Account
Register to borrow books, browse stock, & access digital loans
Username *
e.g. jsmith
Institutional Email *
jsmith@vit.edu
Must contain characters '@' and '.' (e.g. name@vit.edu)
Credential & Password Requirements:
Email must compulsory contain '@' and '.'
Password length: 8 to 64 characters
At least 1 uppercase letter (A-Z)
At least 1 lowercase letter (a-z)
At least 1 numeric digit (0-9)
At least 1 special character (e.g. ! @ # $ % ^ & *)
Password *
Account Type (Role) *
Student (Academic Patron)
Complete Registration →

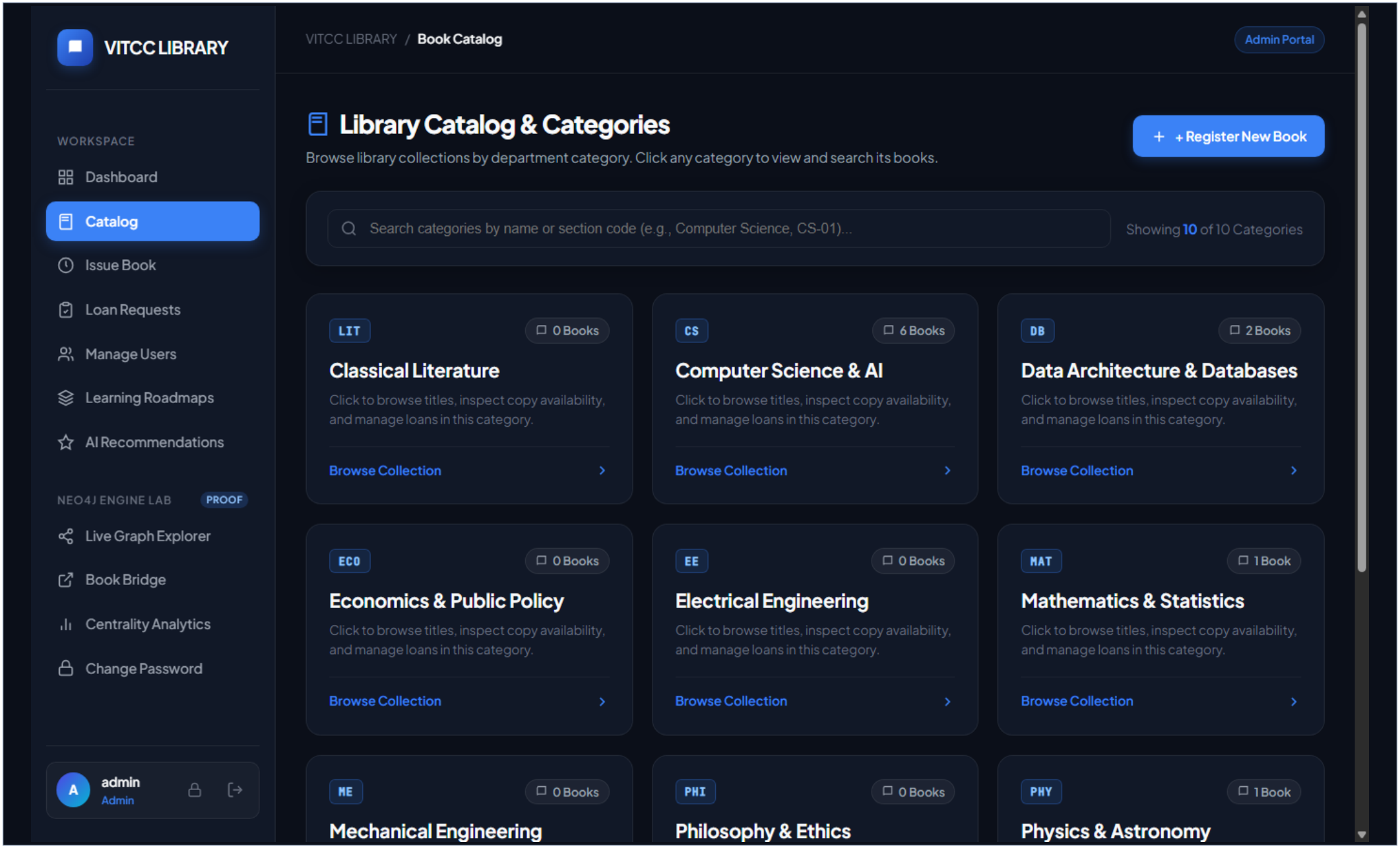
Executive Dashboard & System Metrics

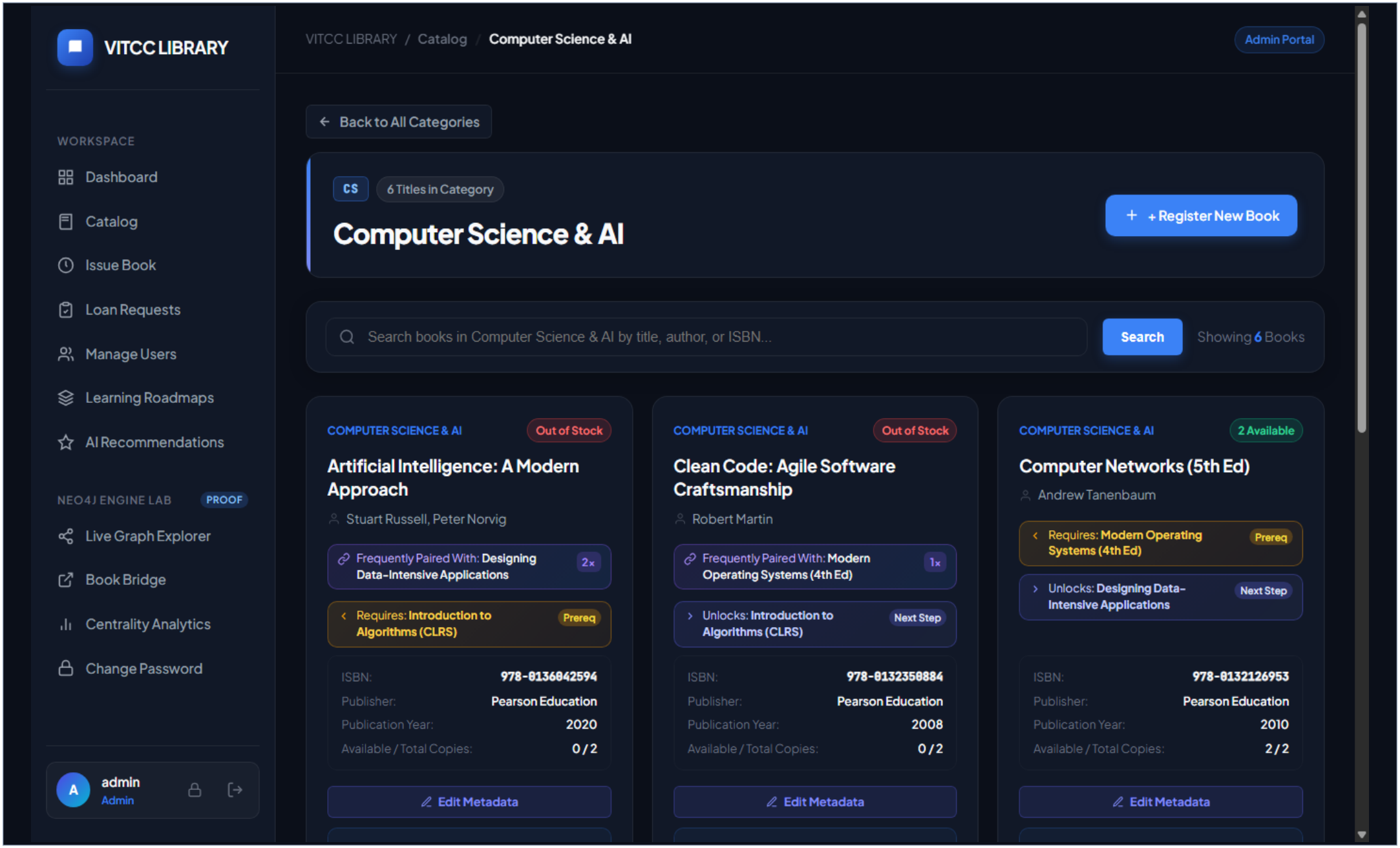
● ADMIN ACCESS

Route: http://127.0.0.1:5000/dashboard | **ID:** 03_admin_dashboard

VITCC Library UI Audit




VITCC LIBRARY
VITCC LIBRARY / Book Catalog
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NEO4J ENGINE LAB
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Centrality Analytics
Change Password
admin
Admin
Library Catalog & Categories
Browse library collections by department category. Click any category to view and search its books.
+ Register New Book
Search categories by name or section code (e.g., Computer Science, CS-01)...
Showing 10 of 10 Categories
LIT
0 Books
Classical Literature
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
CS
6 Books
Computer Science & AI
Click to browse titles, inspect copy availability, and manage loans in this category.
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DB
2 Books
Data Architecture & Databases
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
ECO
0 Books
Economics & Public Policy
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
EE
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Electrical Engineering
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Browse Collection
MAT
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Mathematics & Statistics
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
ME
0 Books
Mechanical Engineering
PHI
0 Books
Philosophy & Ethics
PHY
1 Book
Physics & Astronomy

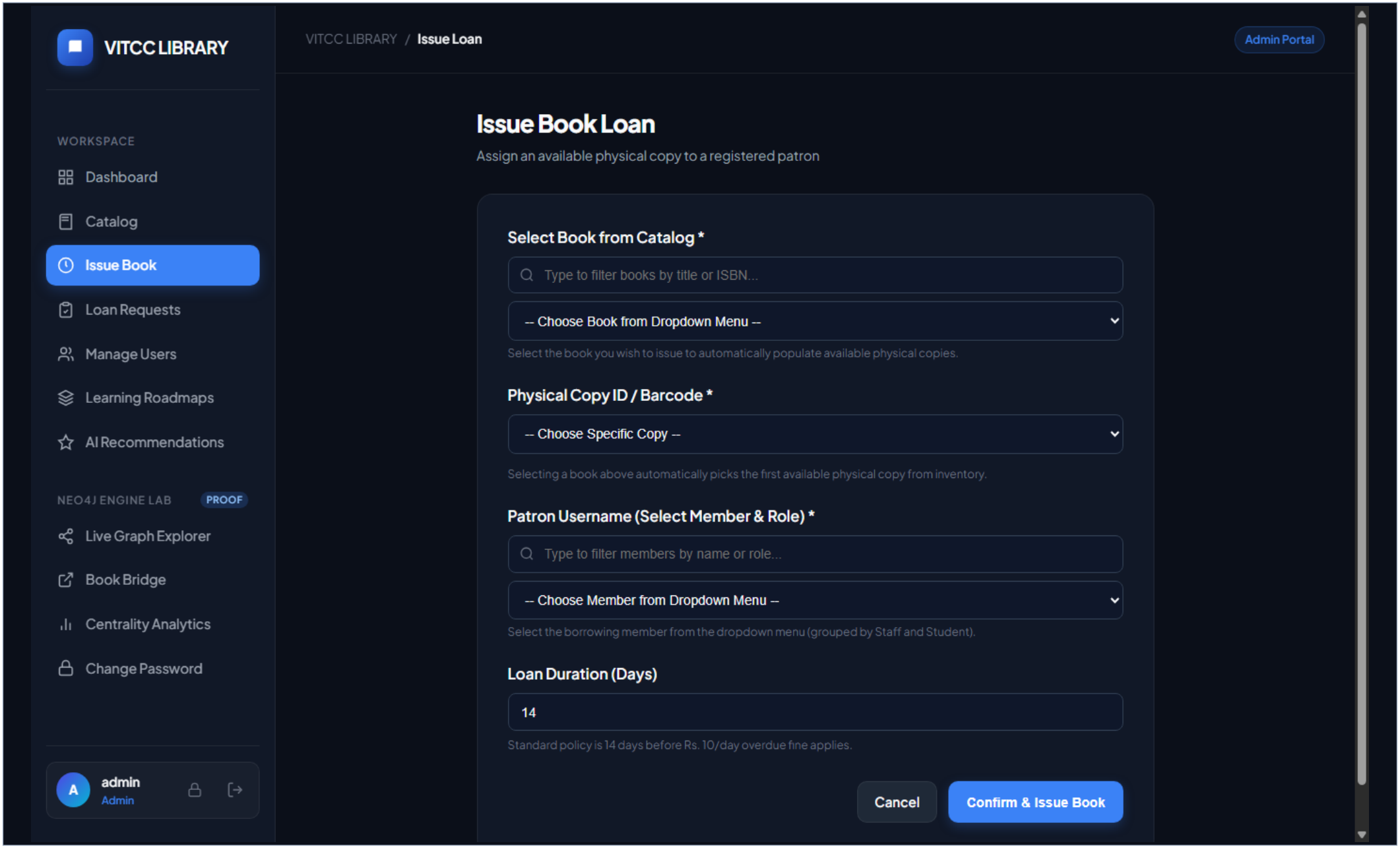

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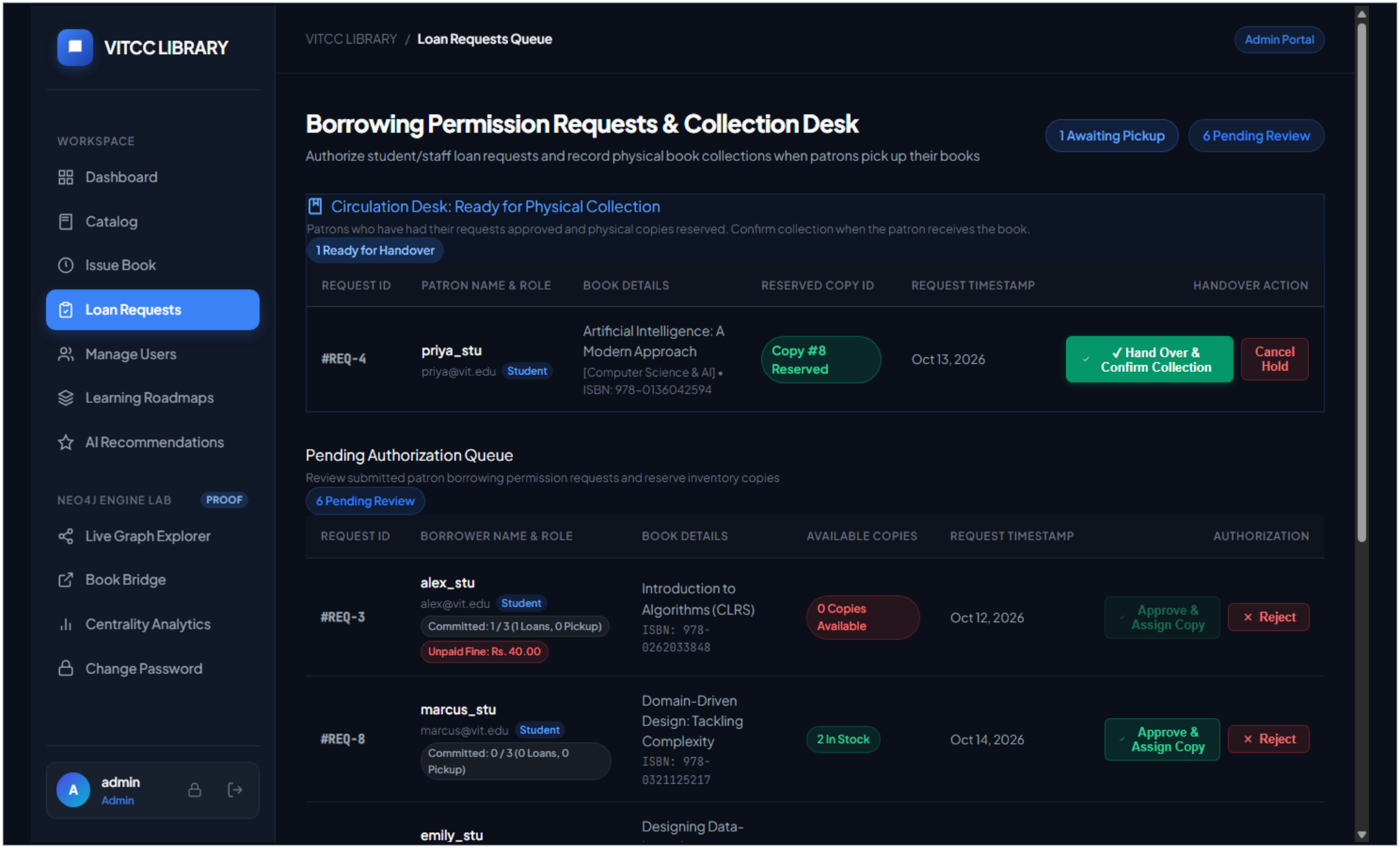
Circulation Desk - Direct Book Issuance

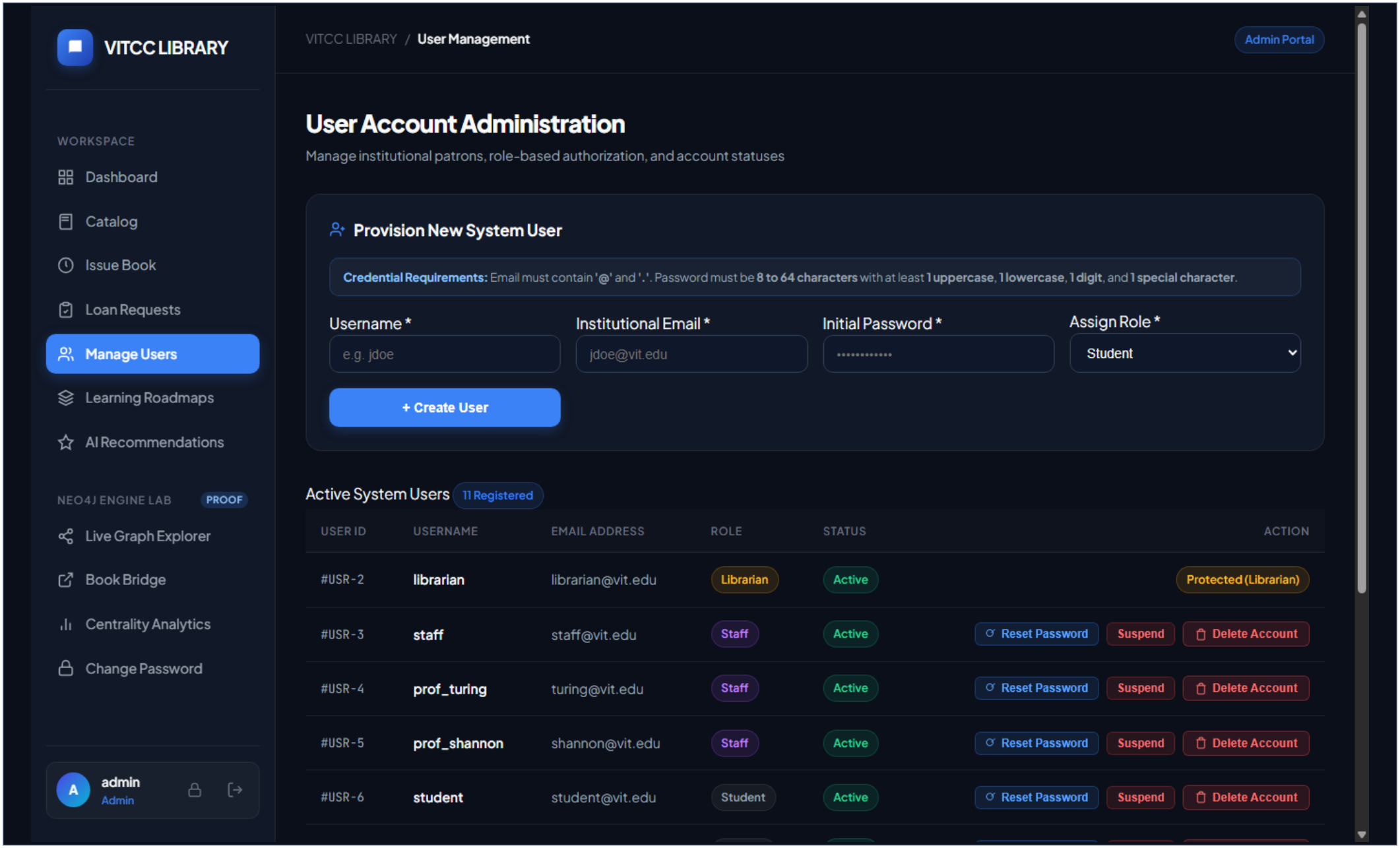
● ADMIN ACCESS

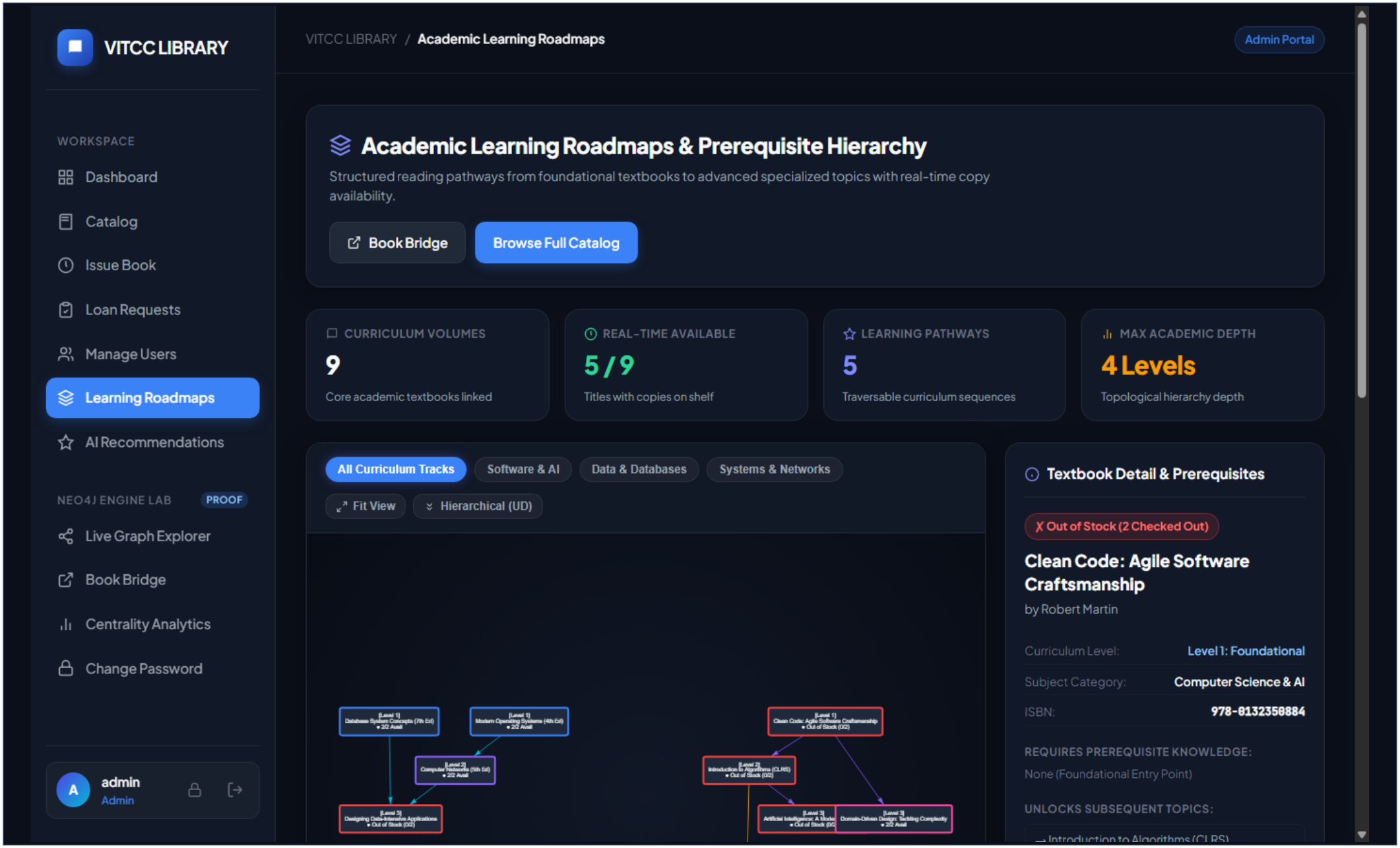
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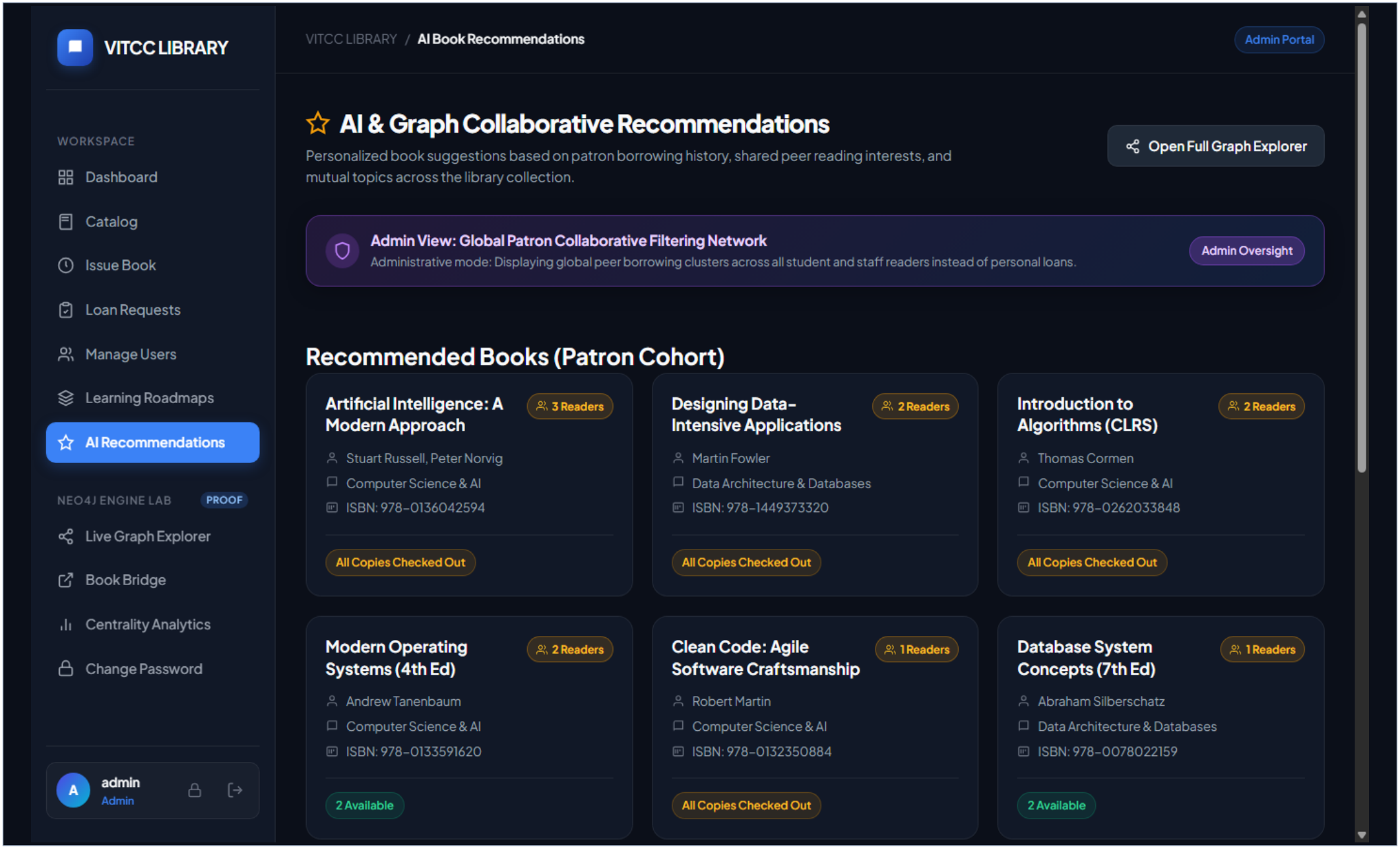
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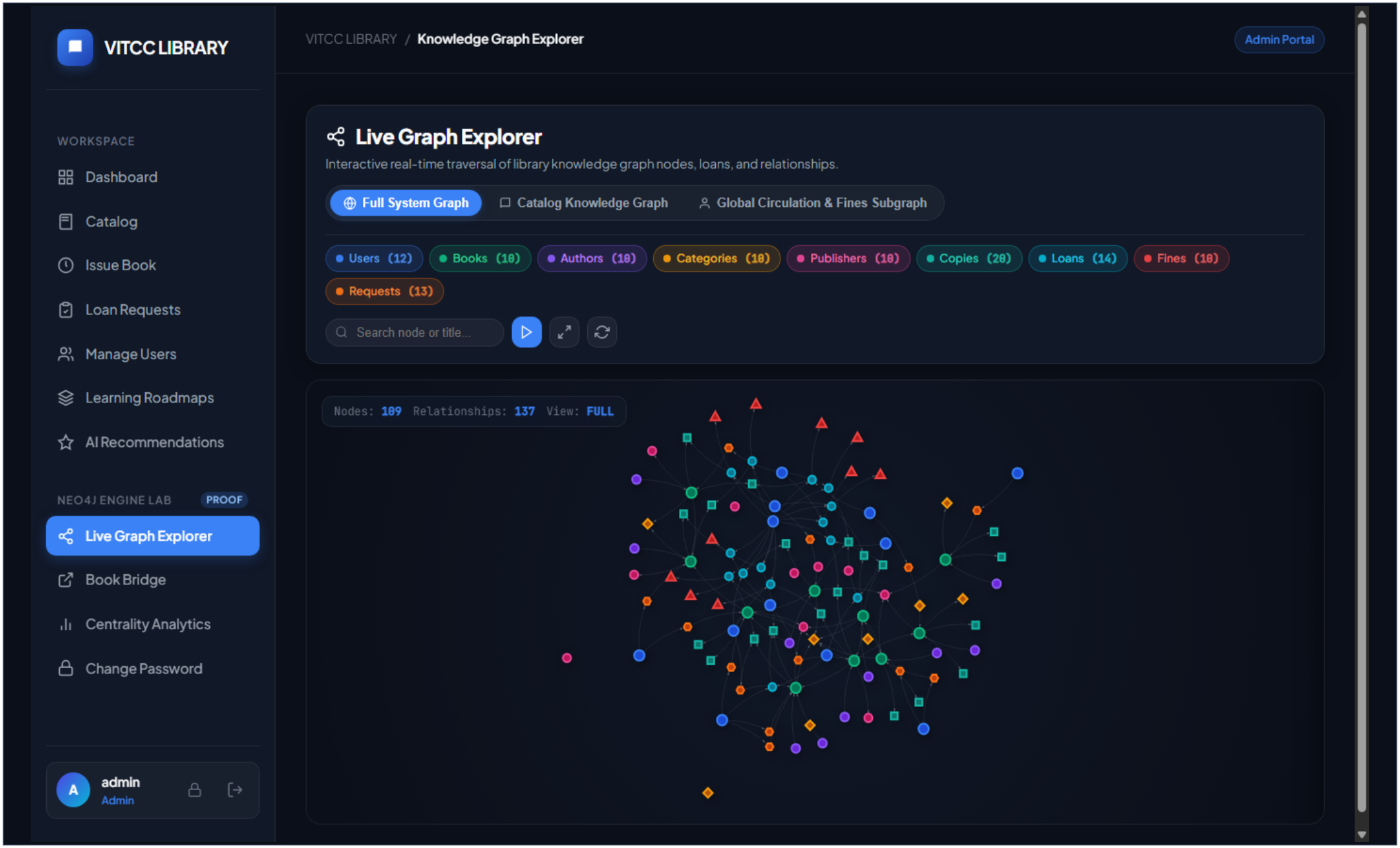



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VITCC LIBRARY / Loan Requests Queue
Admin Portal
Borrowing Permission Requests & Collection Desk
Authorize student/staff loan requests and record physical book collections when patrons pick up their books
1 Awaiting Pickup
6 Pending Review
Circulation Desk: Ready for Physical Collection
Patrons who have had their requests approved and physical copies reserved. Confirm collection when the patron receives the book.
1 Ready for Handover
REQUEST ID
PATRON NAME & ROLE
BOOK DETAILS
RESERVED COPY ID
REQUEST TIMESTAMP
HANDOVER ACTION
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admin
Admin
User Account Administration
Manage institutional patrons, role-based authorization, and account statuses
Provision New System User
Credential Requirements: Email must contain '@' and '.'. Password must be 8 to 64 characters with at least 1 uppercase, 1 lowercase, 1 digit, and 1 special character.
Username *
e.g. jdoe
Institutional Email *
jdoe@vit.edu
Initial Password *
Assign Role *
Student
+ Create User
Active System Users
11 Registered
USER ID
USERNAME
EMAIL ADDRESS
ROLE
STATUS
ACTION
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librarian@vit.edu
Librarian
Active
Protected (Librarian)
#USR-3
staff
staff@vit.edu
Staff
Active
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Suspend
Delete Account
#USR-4
prof_turing
turing@vit.edu
Staff
Active
Reset Password
Suspend
Delete Account
#USR-5
prof_shannon
shannon@vit.edu
Staff
Active
Reset Password
Suspend
Delete Account
#USR-6
student
student@vit.edu
Student
Active
Reset Password
Suspend
Delete Account


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Systems & Networks
Fit View
Hierarchical (UD)
Textbook Detail & Prerequisites
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Clean Code: Agile Software Craftsmanship
by Robert Martin
Curriculum Level:
Level 1: Foundational
Subject Category:
Computer Science & AI
ISBN:
978-0132350884
REQUIRES PREREQUISITE KNOWLEDGE:
None (Foundational Entry Point)
UNLOCKS SUBSEQUENT TOPICS:


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AI & Graph Collaborative Recommendations
Personalized book suggestions based on patron borrowing history, shared peer reading interests, and mutual topics across the library collection.
Open Full Graph Explorer
Admin View: Global Patron Collaborative Filtering Network
Administrative mode: Displaying global peer borrowing clusters across all student and staff readers instead of personal loans.
Admin Oversight
Recommended Books (Patron Cohort)
Artificial Intelligence: A Modern Approach
3 Readers
Stuart Russell, Peter Norvig
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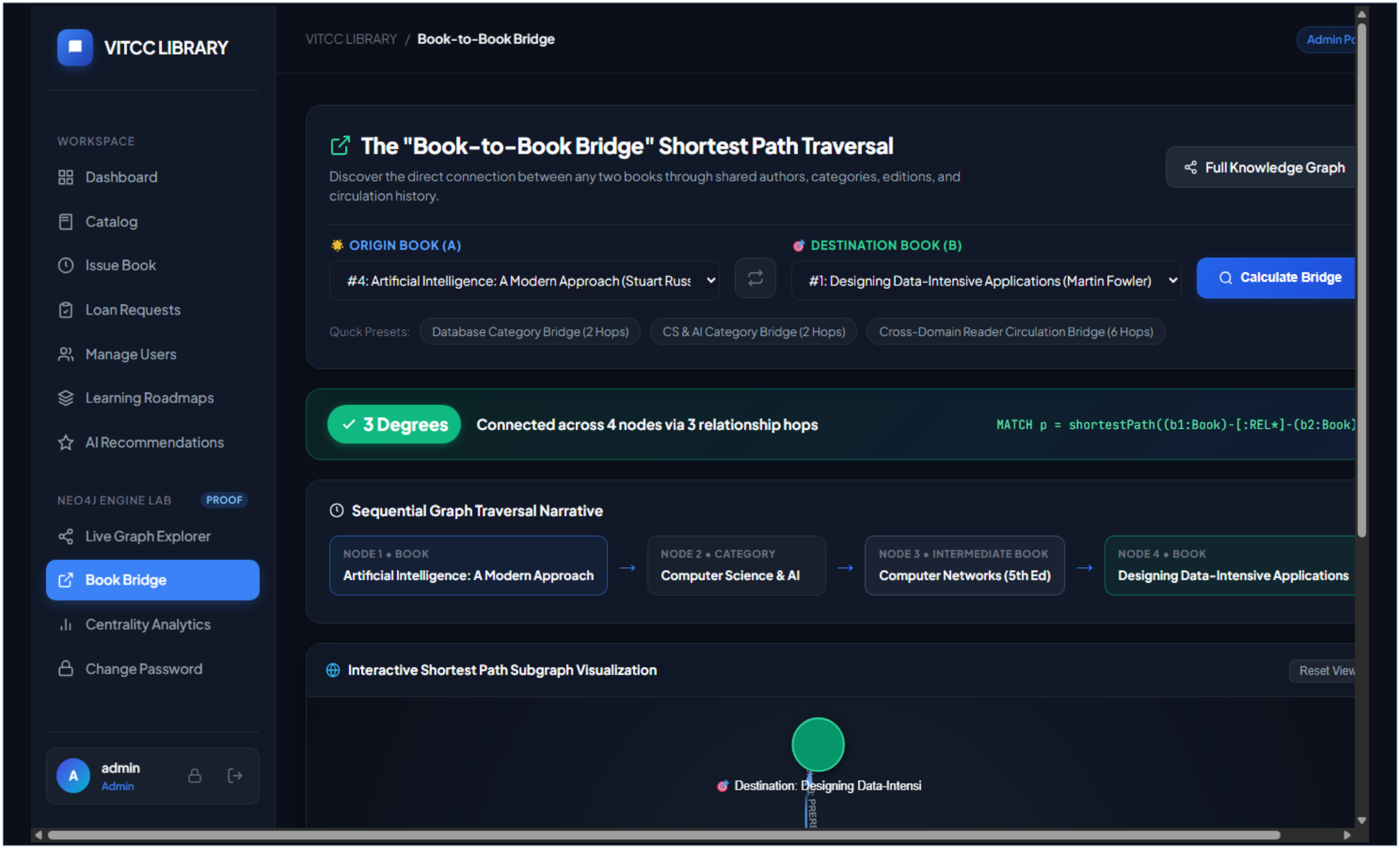

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Admin
Live Graph Explorer
Interactive real-time traversal of library knowledge graph nodes, loans, and relationships.
Full System Graph
Catalog Knowledge Graph
Global Circulation & Fines Subgraph
Users (12)
Books (18)
Authors (18)
Categories (18)
Publishers (18)
Copies (28)
Loans (14)
Fines (18)
Requests (13)
Search node or title...
Nodes: 109 Relationships: 137 View: FULL

Book Bridge - Shortest Path Separation

● ADMIN ACCESS

Route: http://127.0.0.1:5000/book_bridge | **ID:** 12_admin_book_bridge

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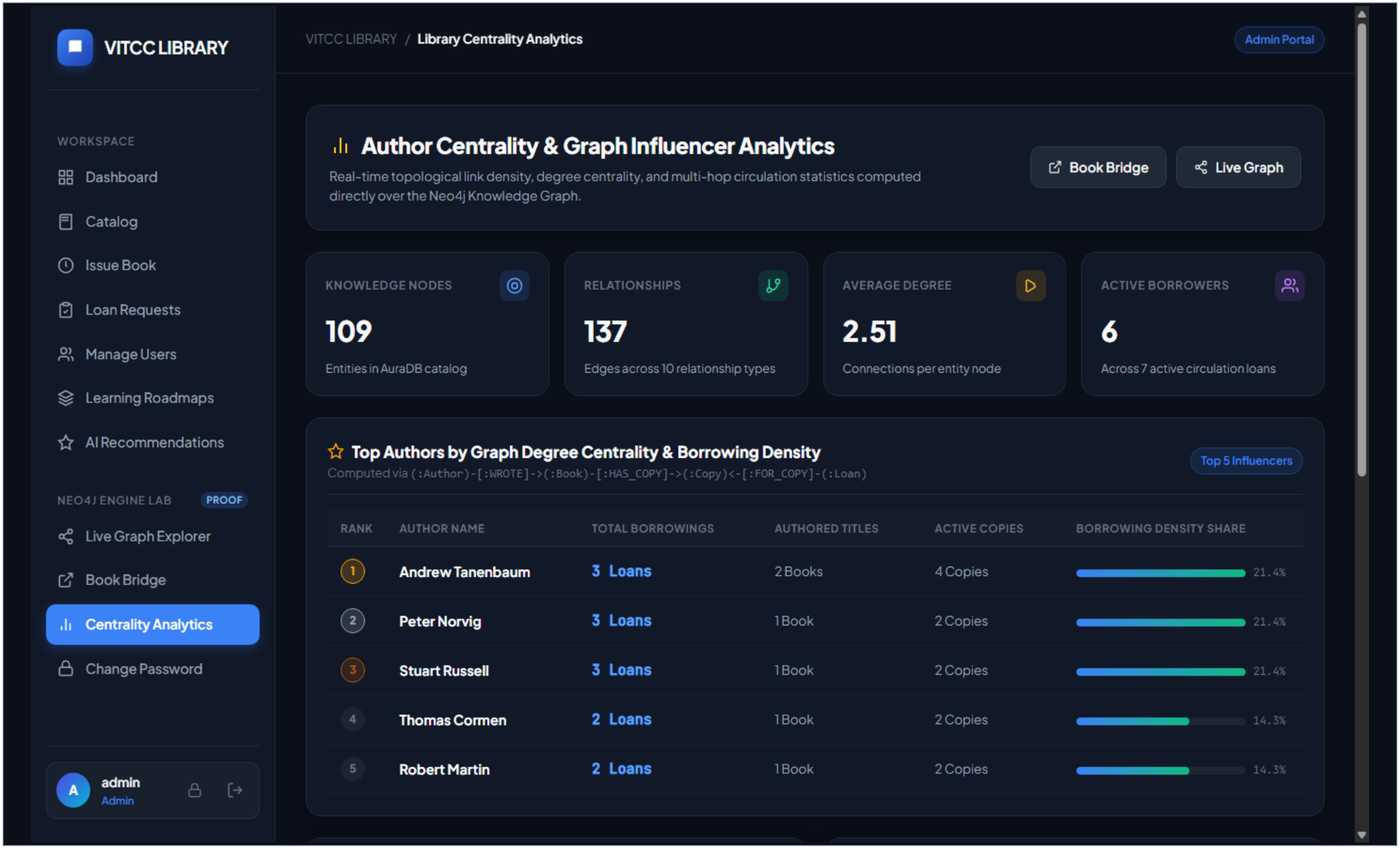


Centrality Analytics & Influencer Lab

● ADMIN ACCESS

Route: http://127.0.0.1:5000/library_analytics | **ID:** 13_admin_centrality_analytics

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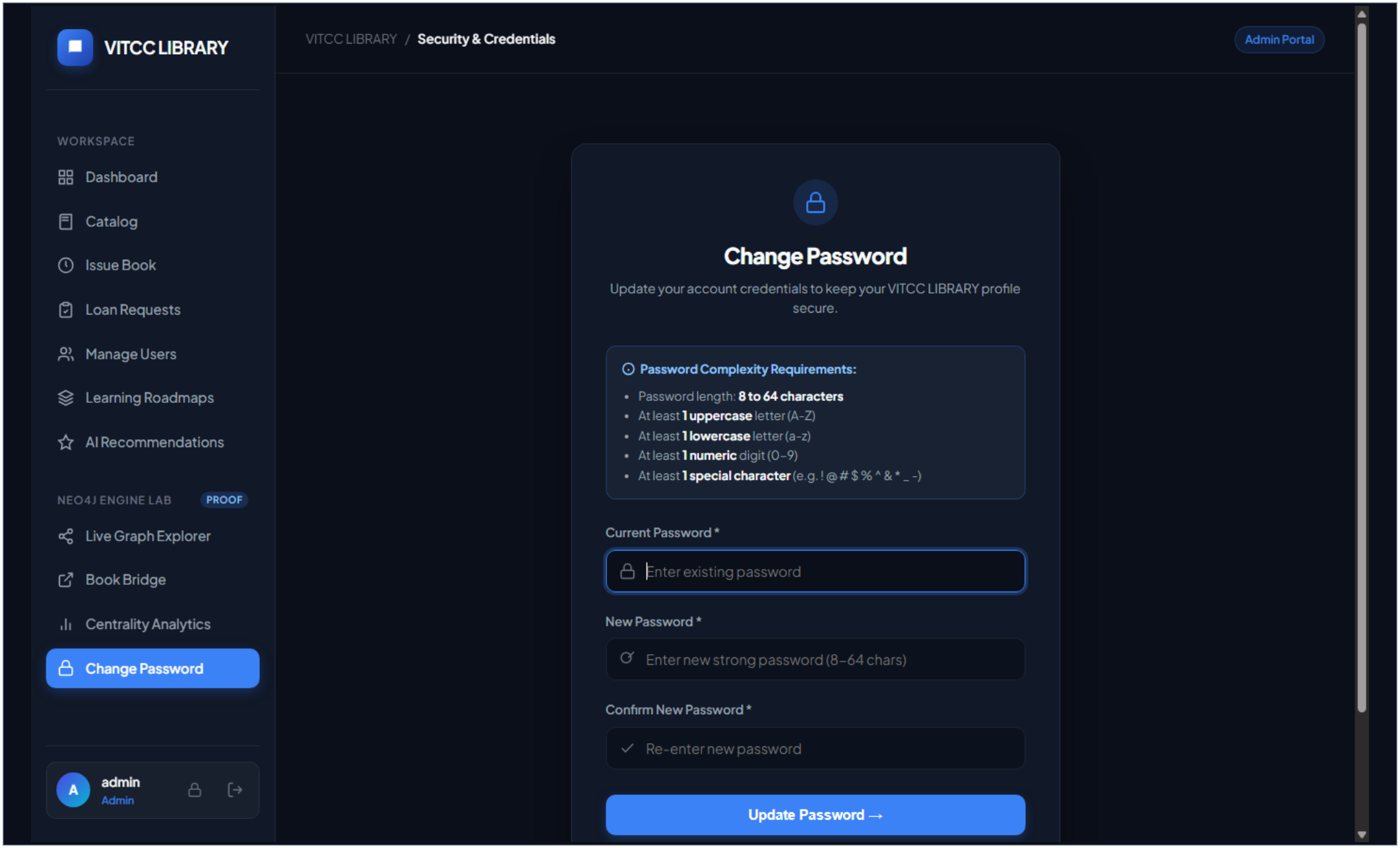


Security Credentials & Password Management

● ADMIN ACCESS

Route: http://127.0.0.1:5000/change_password | **ID:** 14_admin_change_password

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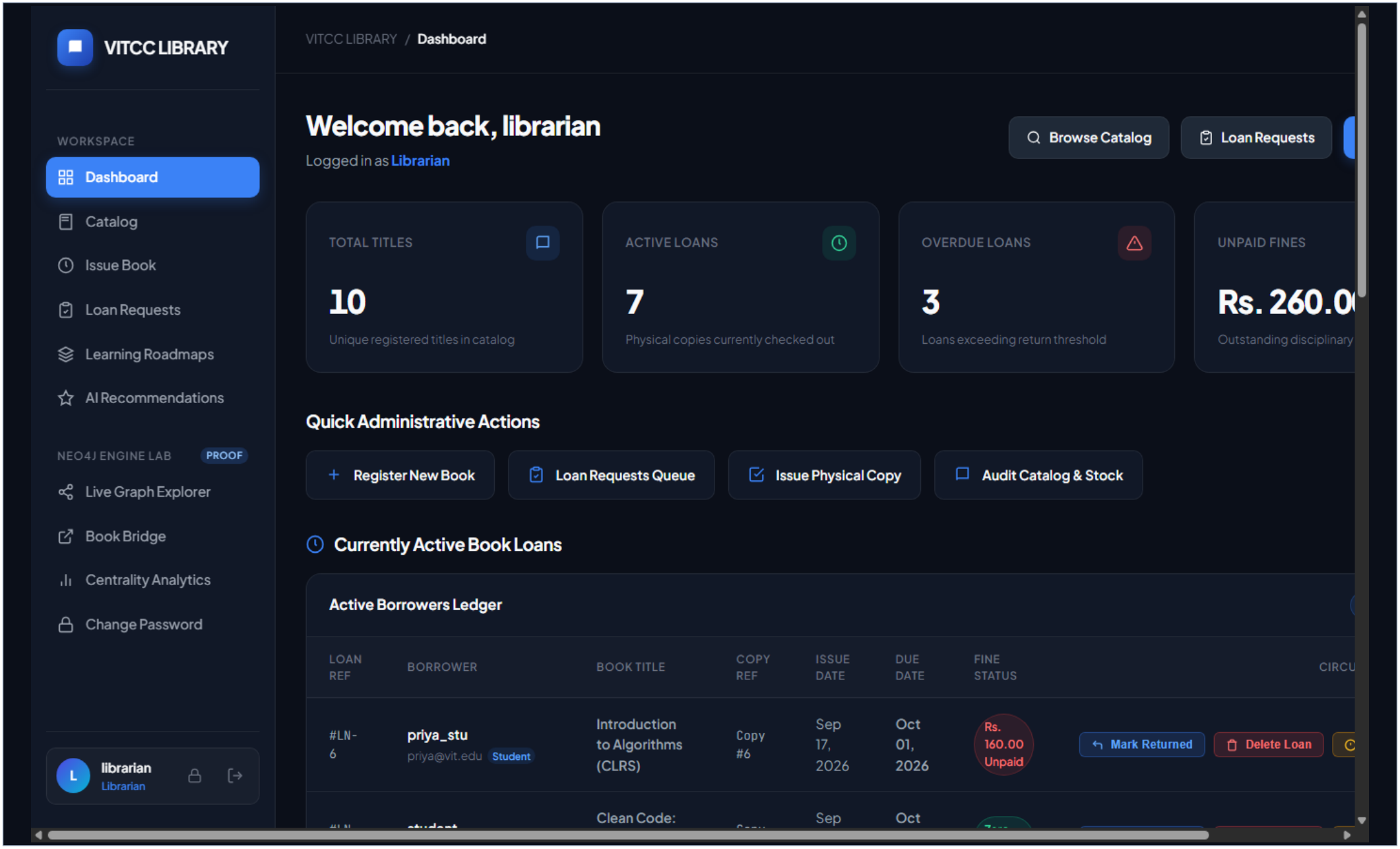


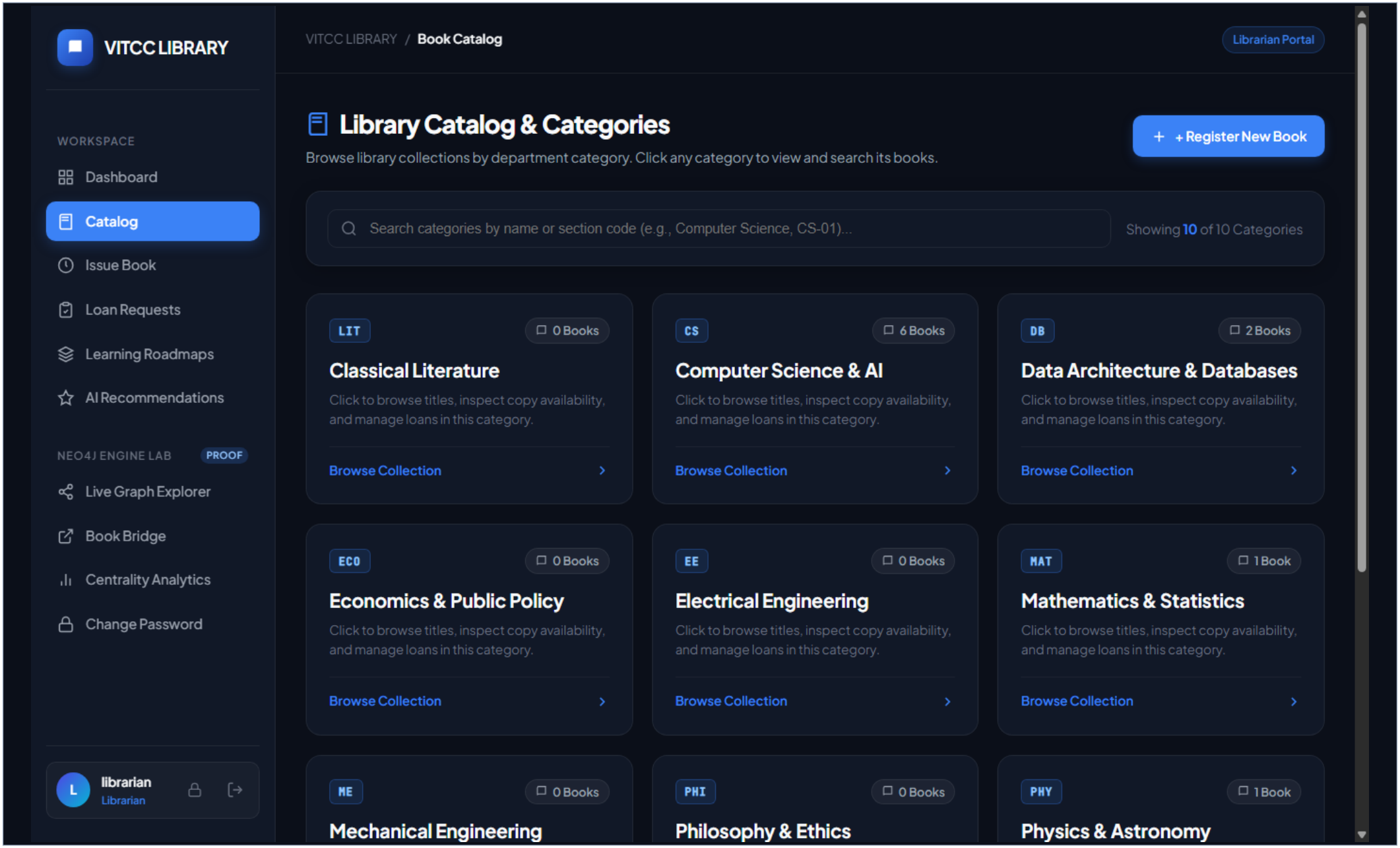
Circulation Desk Dashboard

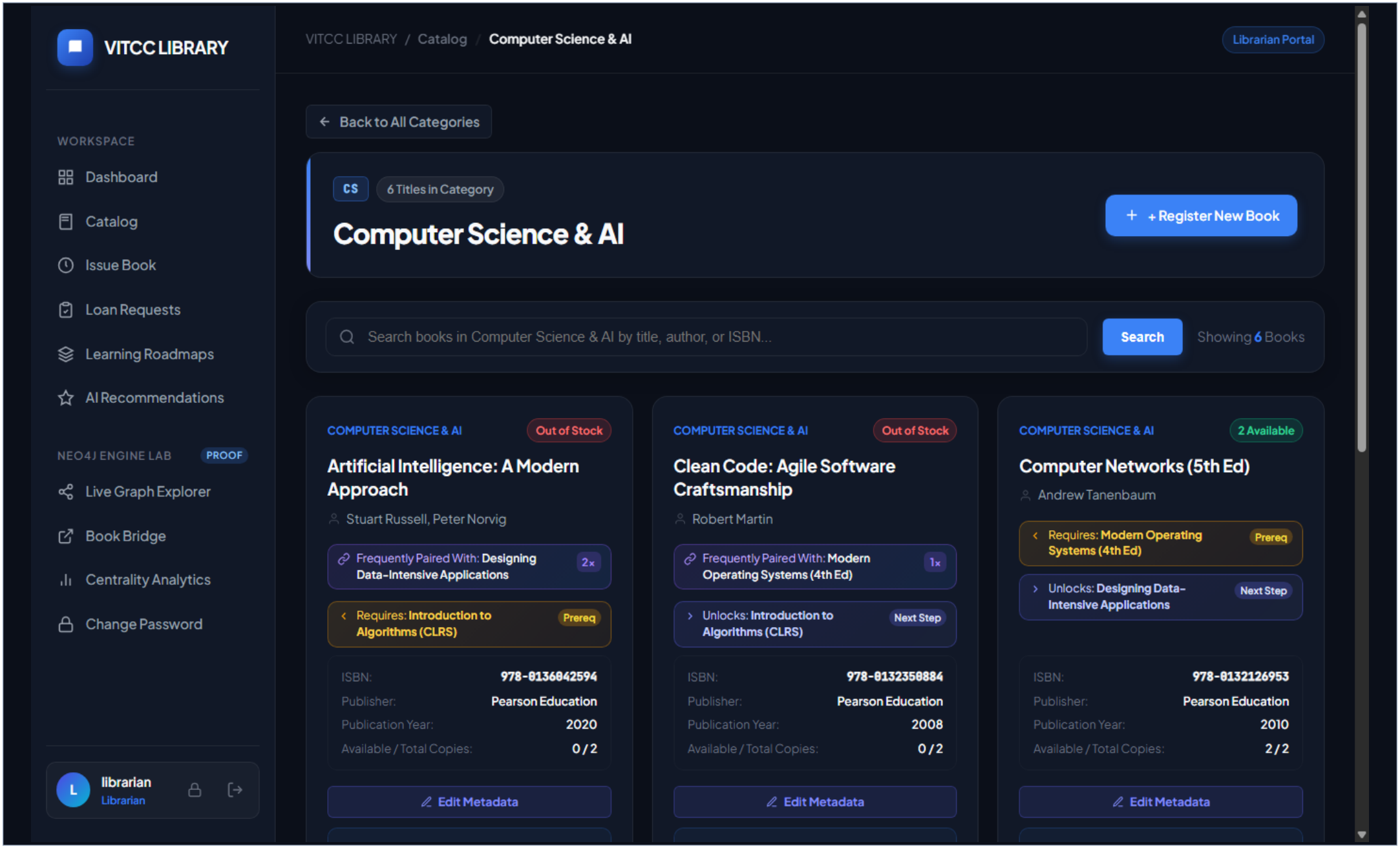
● LIBRARIAN ACCESS

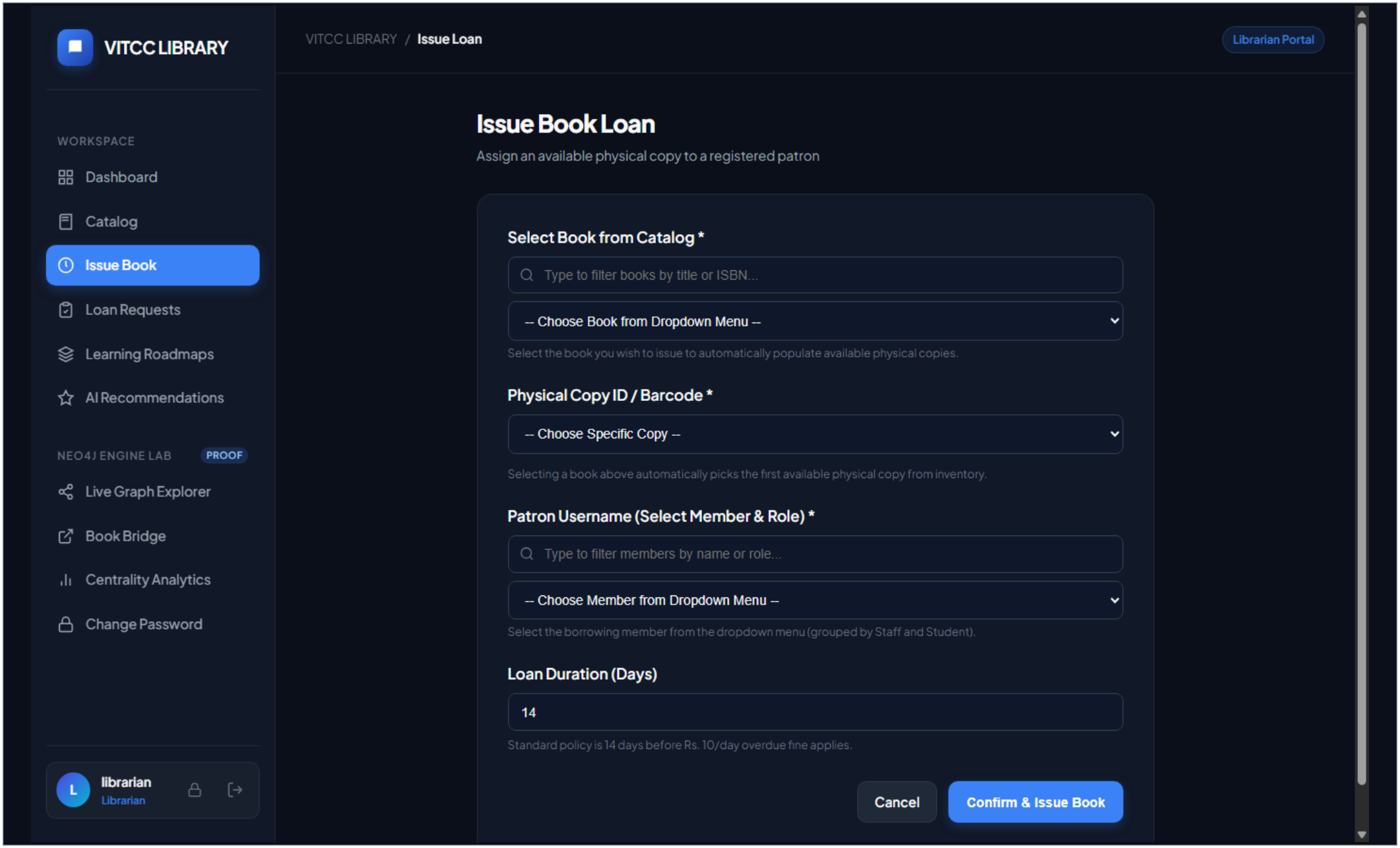
Route: http://127.0.0.1:5000/dashboard | **ID:** 15_librarian_dashboard

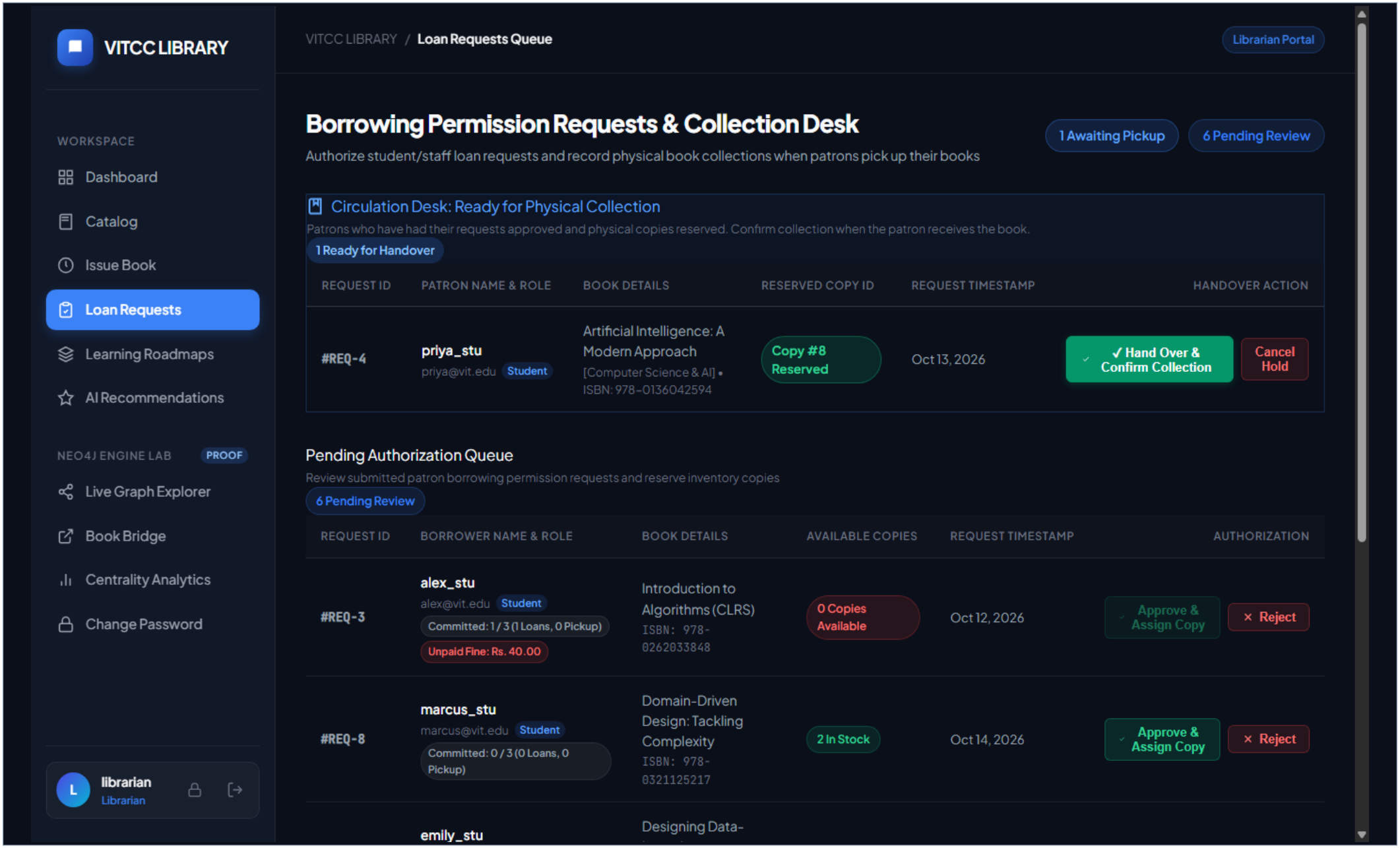
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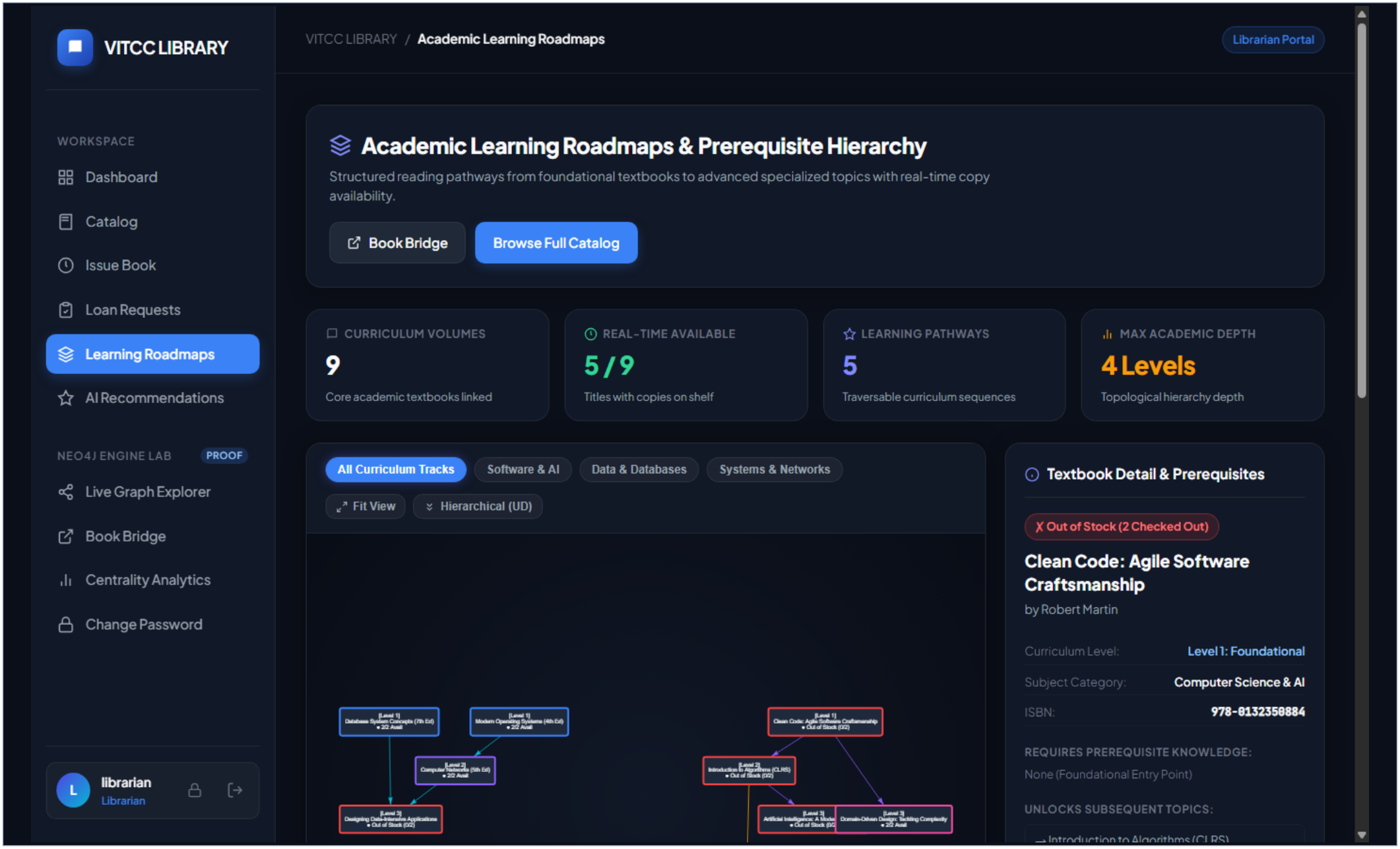


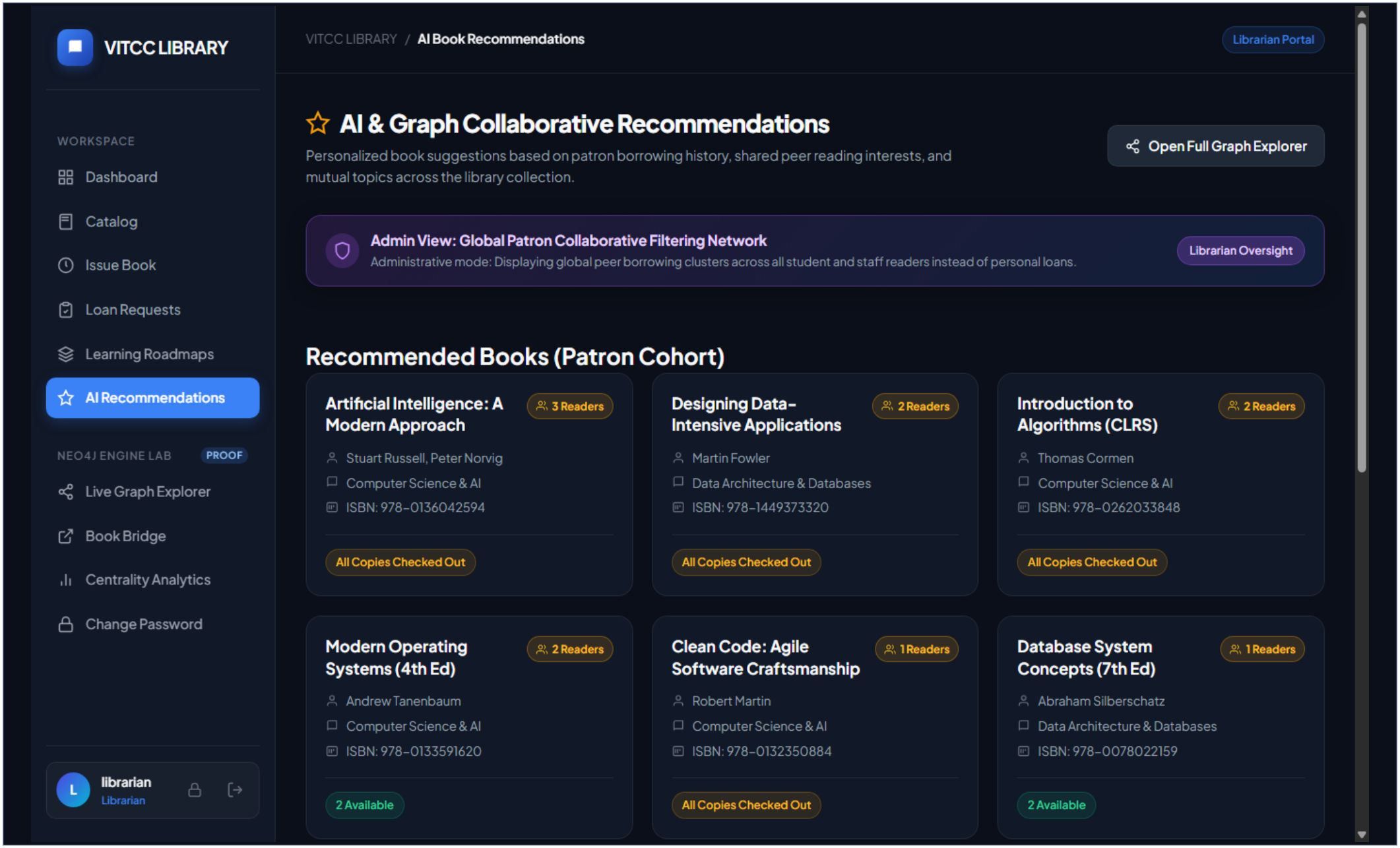

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Click to browse titles, inspect copy availability, and manage loans in this category.
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CS
6 Books
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Click to browse titles, inspect copy availability, and manage loans in this category.
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DB
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Click to browse titles, inspect copy availability, and manage loans in this category.
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ECO
0 Books
Economics & Public Policy
Click to browse titles, inspect copy availability, and manage loans in this category.
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1 Book
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Click to browse titles, inspect copy availability, and manage loans in this category.
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PHI
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Hierarchical (UD)
Textbook Detail & Prerequisites
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by Robert Martin
Curriculum Level:
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Subject Category:
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ISBN:
978-0132350884
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None (Foundational Entry Point)
UNLOCKS SUBSEQUENT TOPICS:

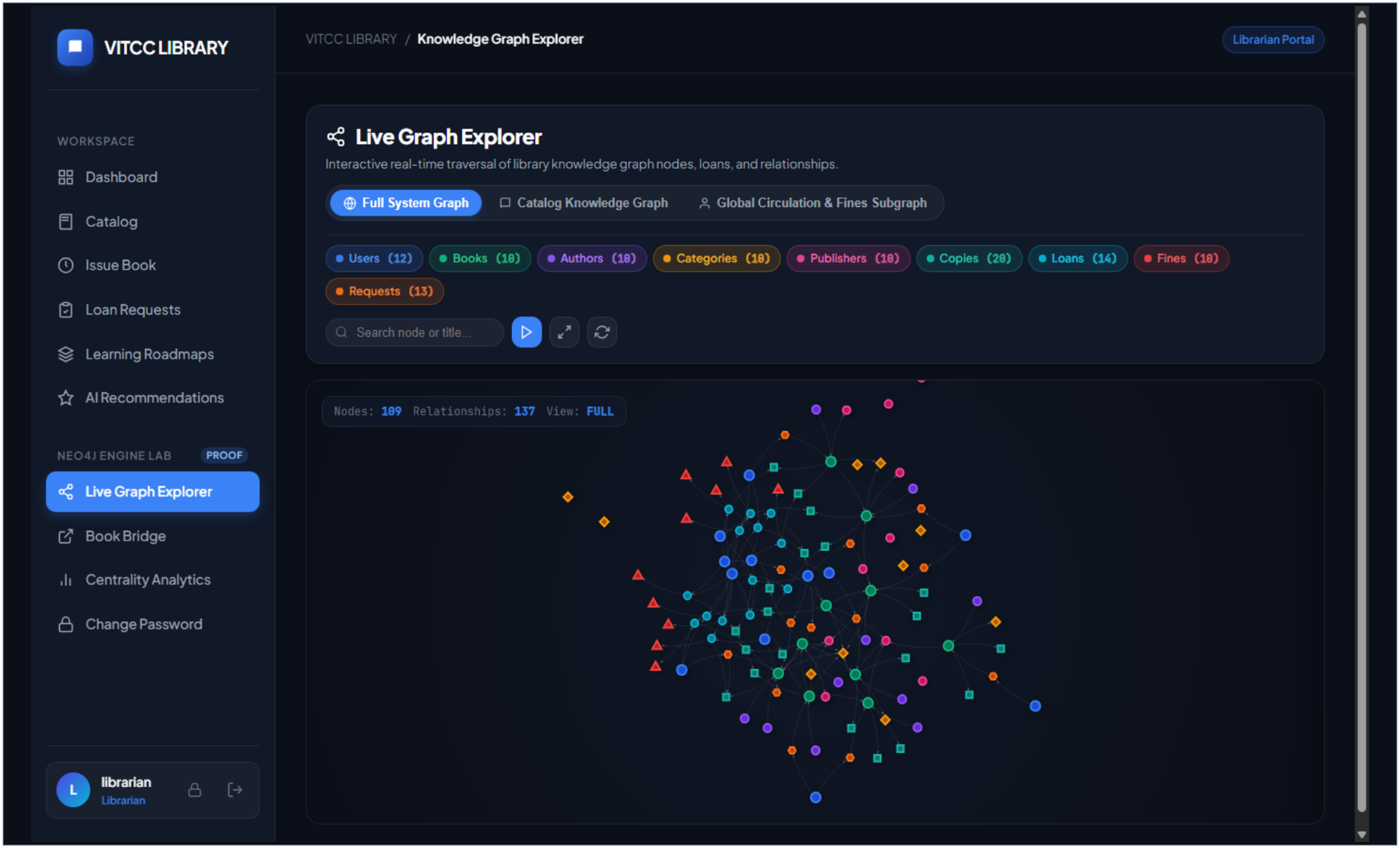

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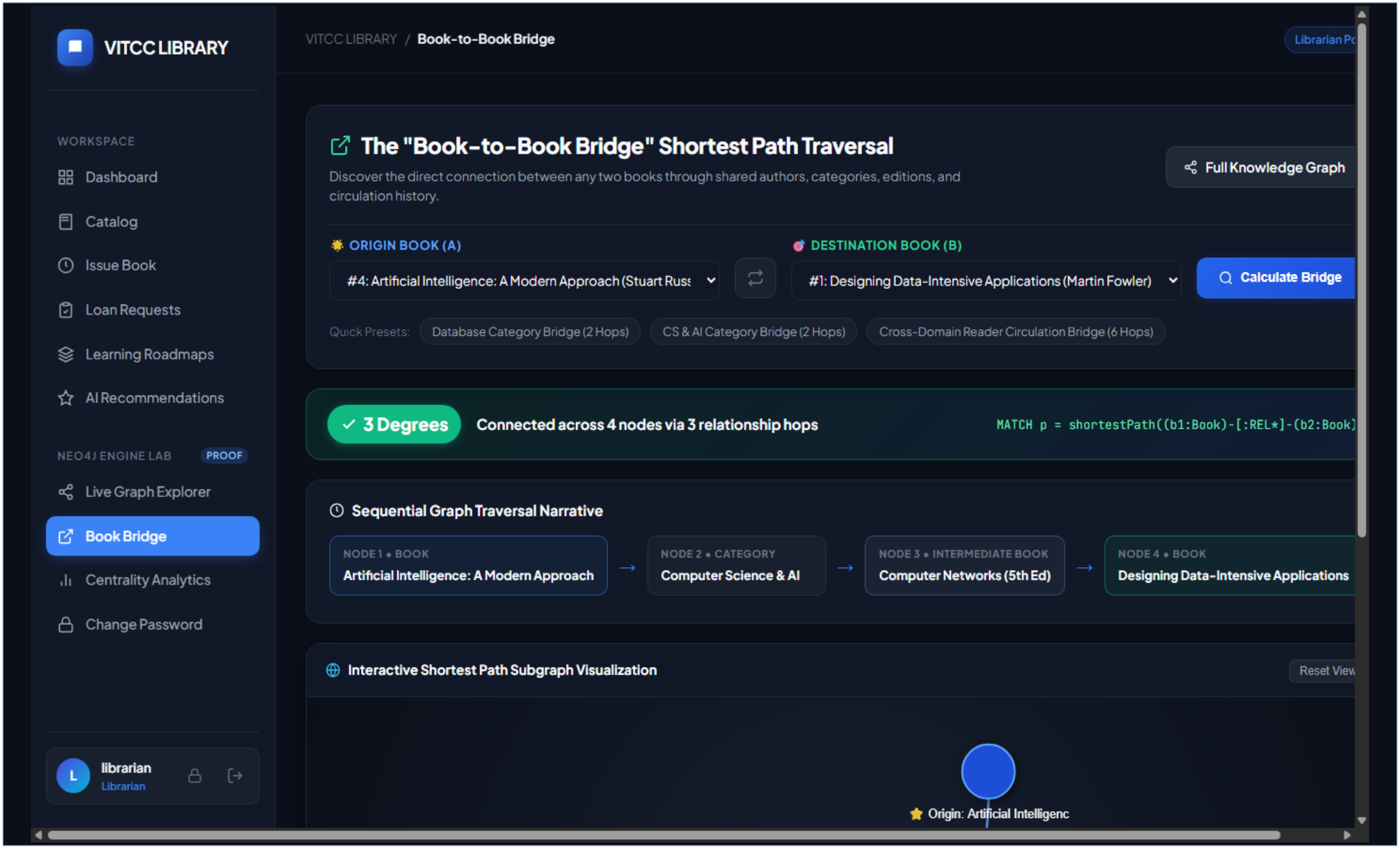
System Circulation Knowledge Graph

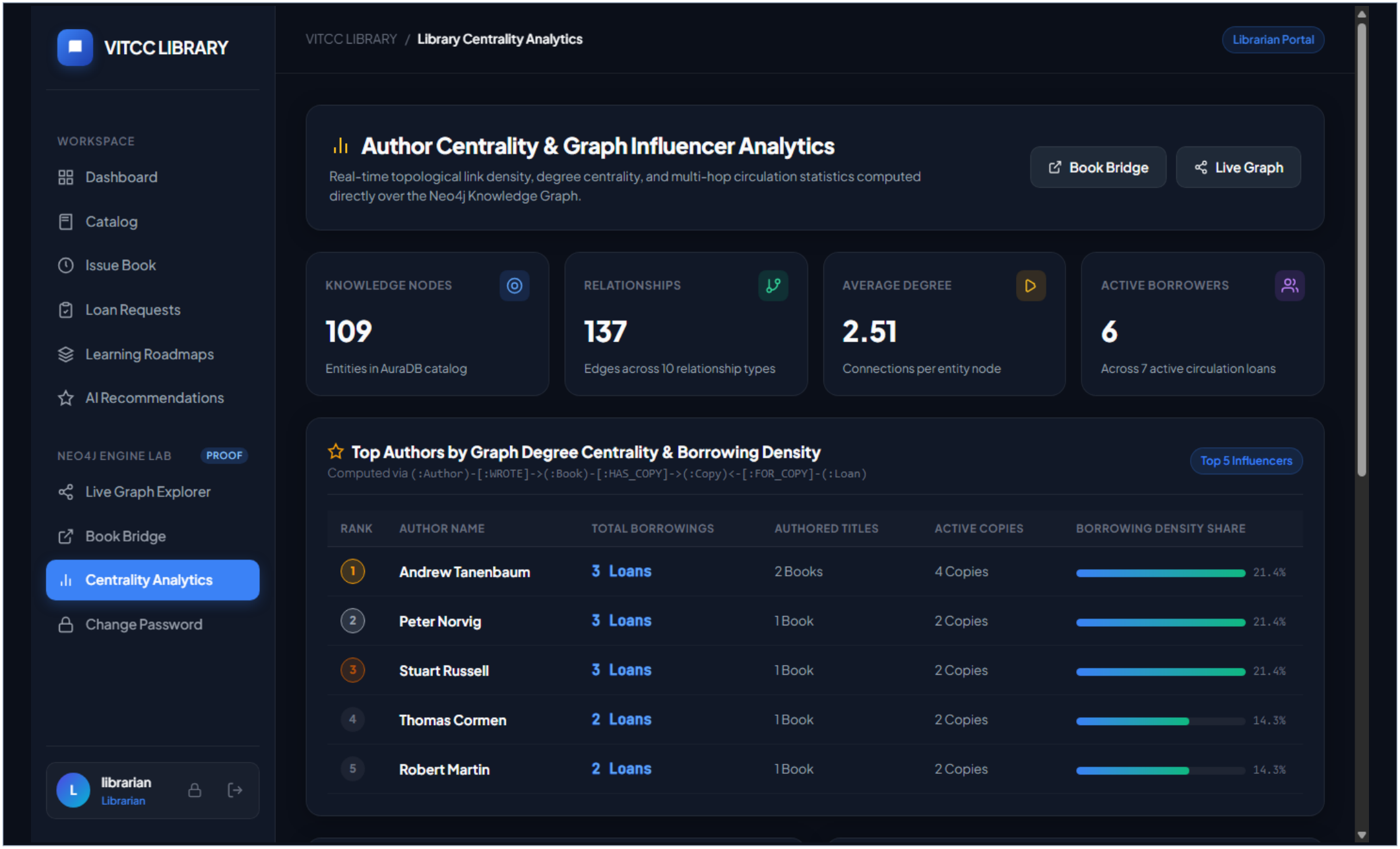
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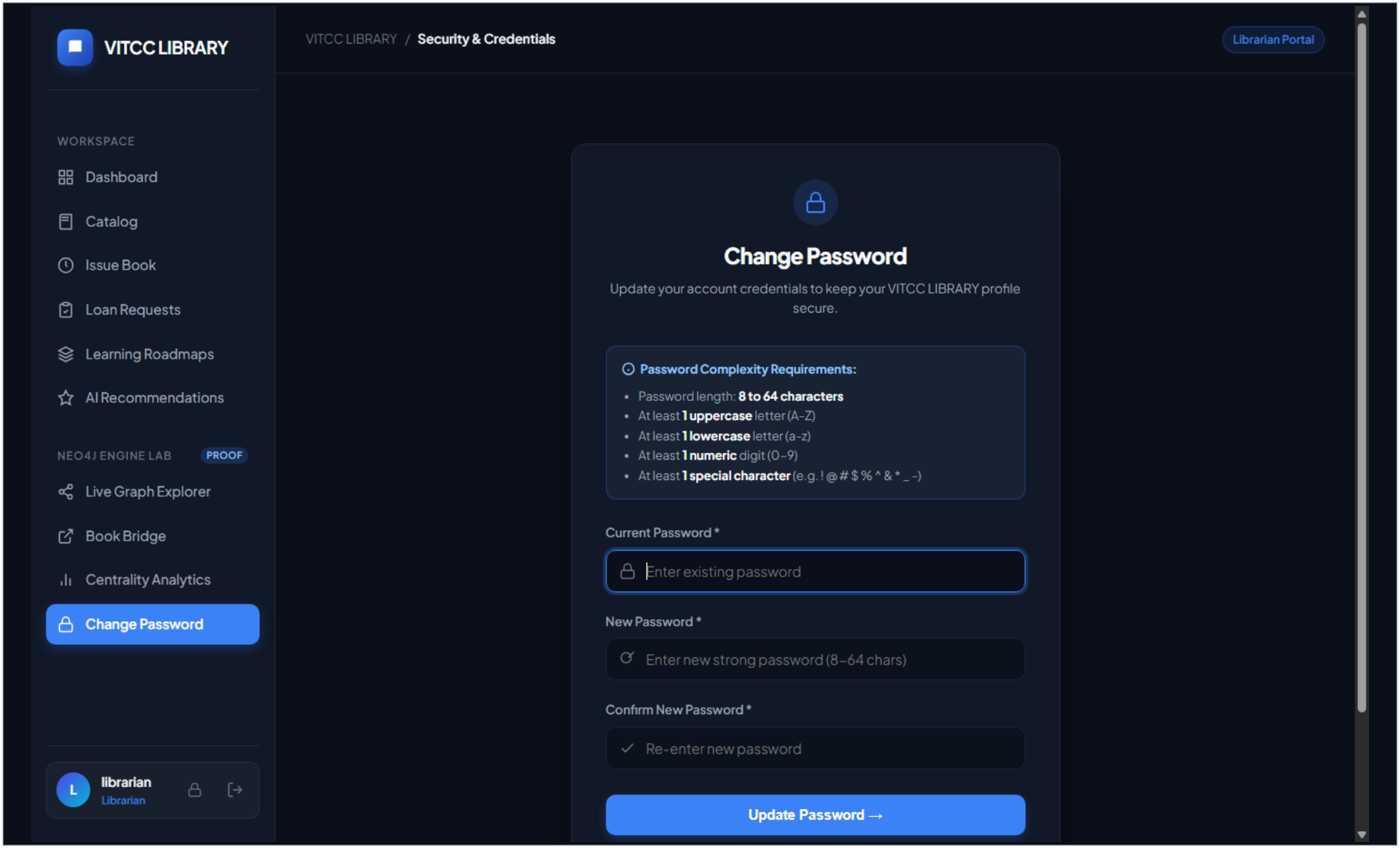
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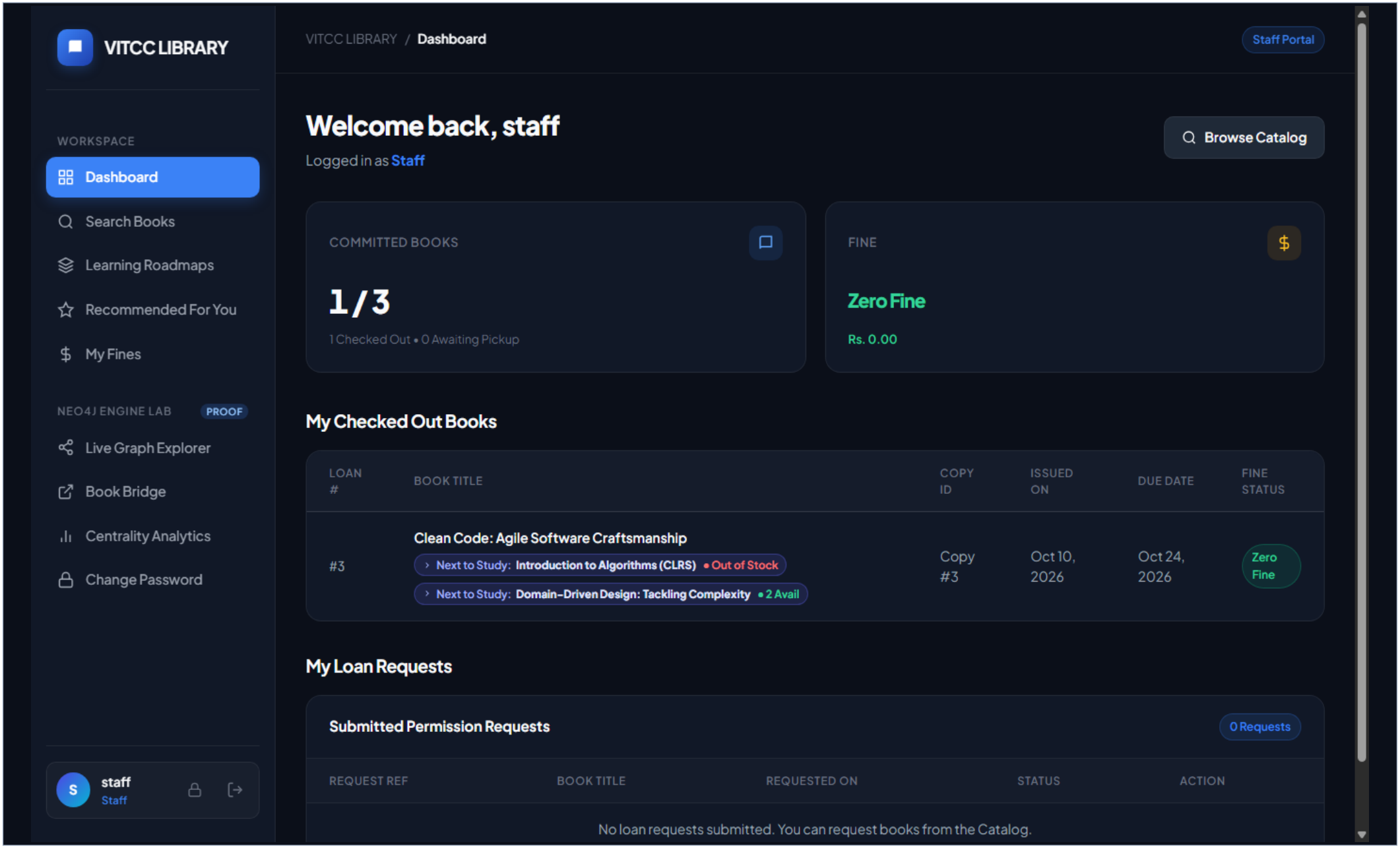
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Connections per entity node
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Across 7 active circulation loans
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New Password *
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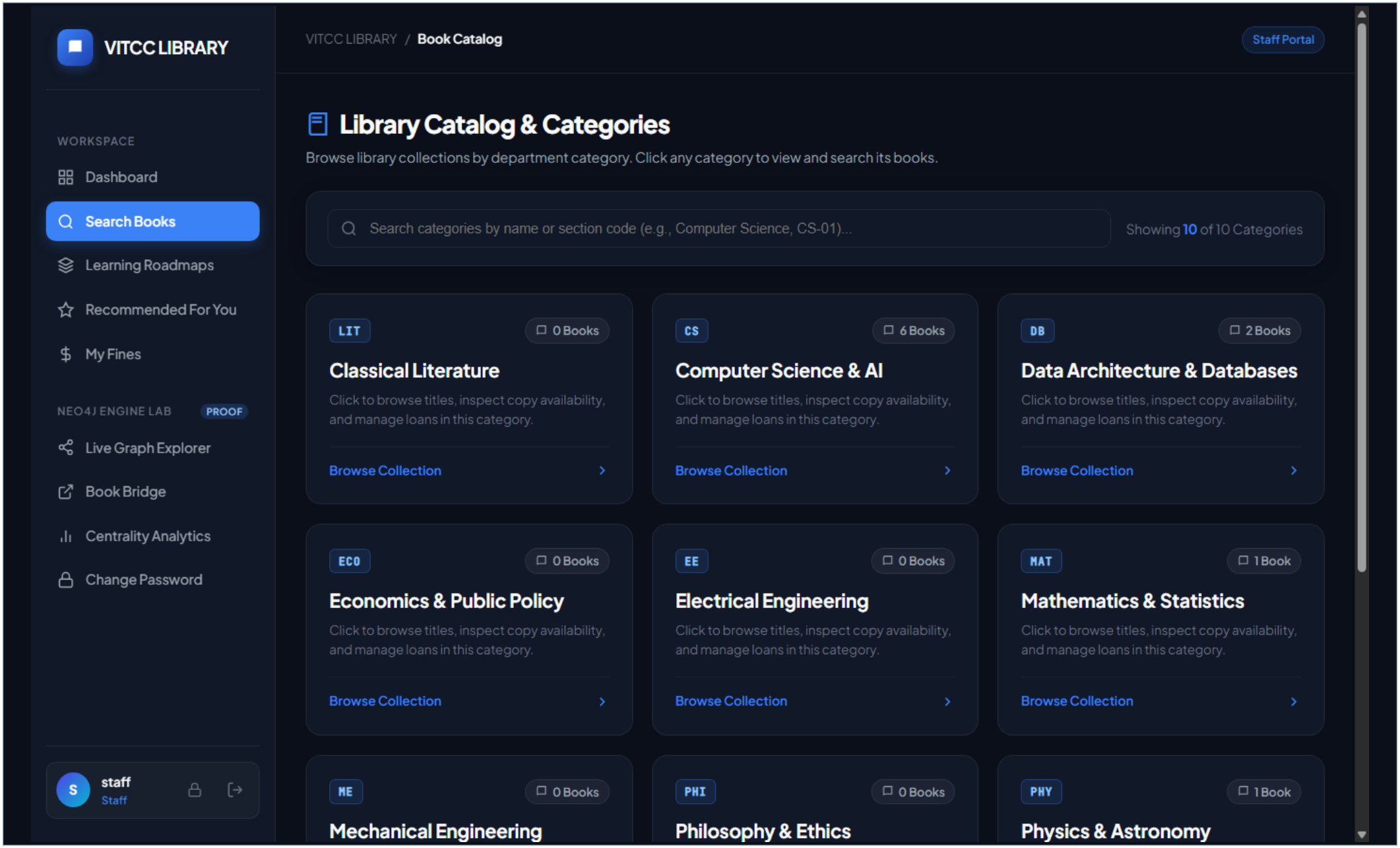

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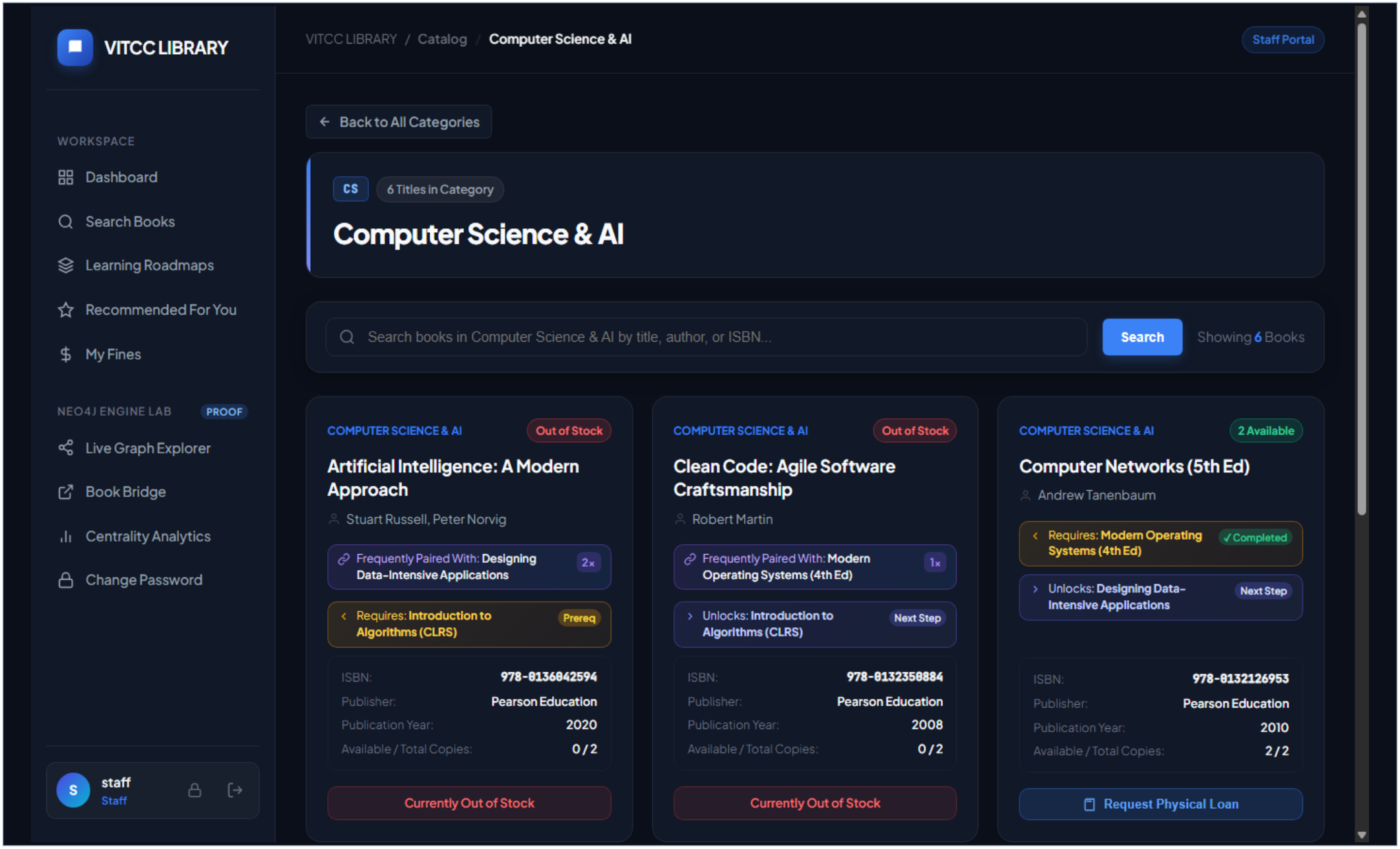
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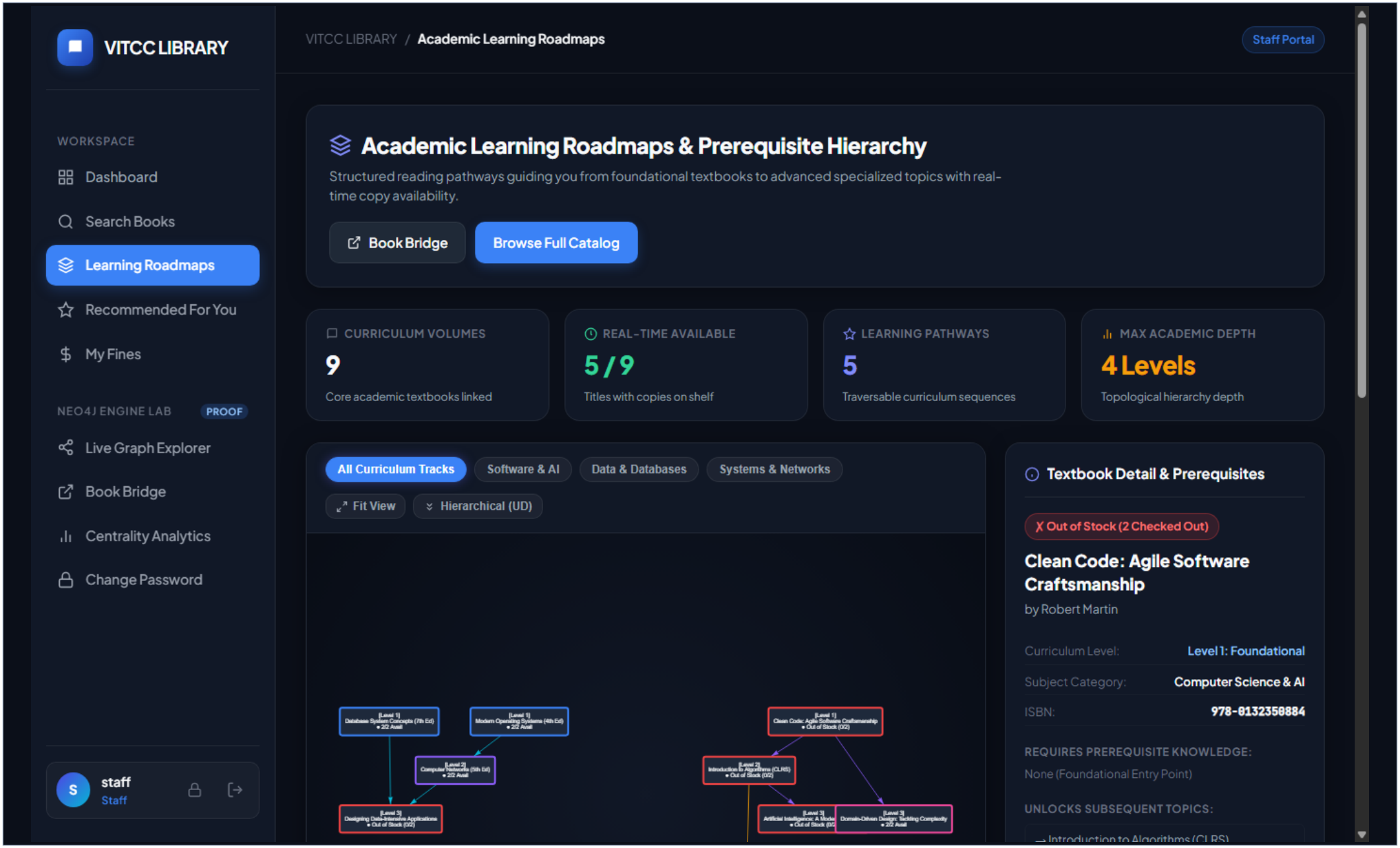
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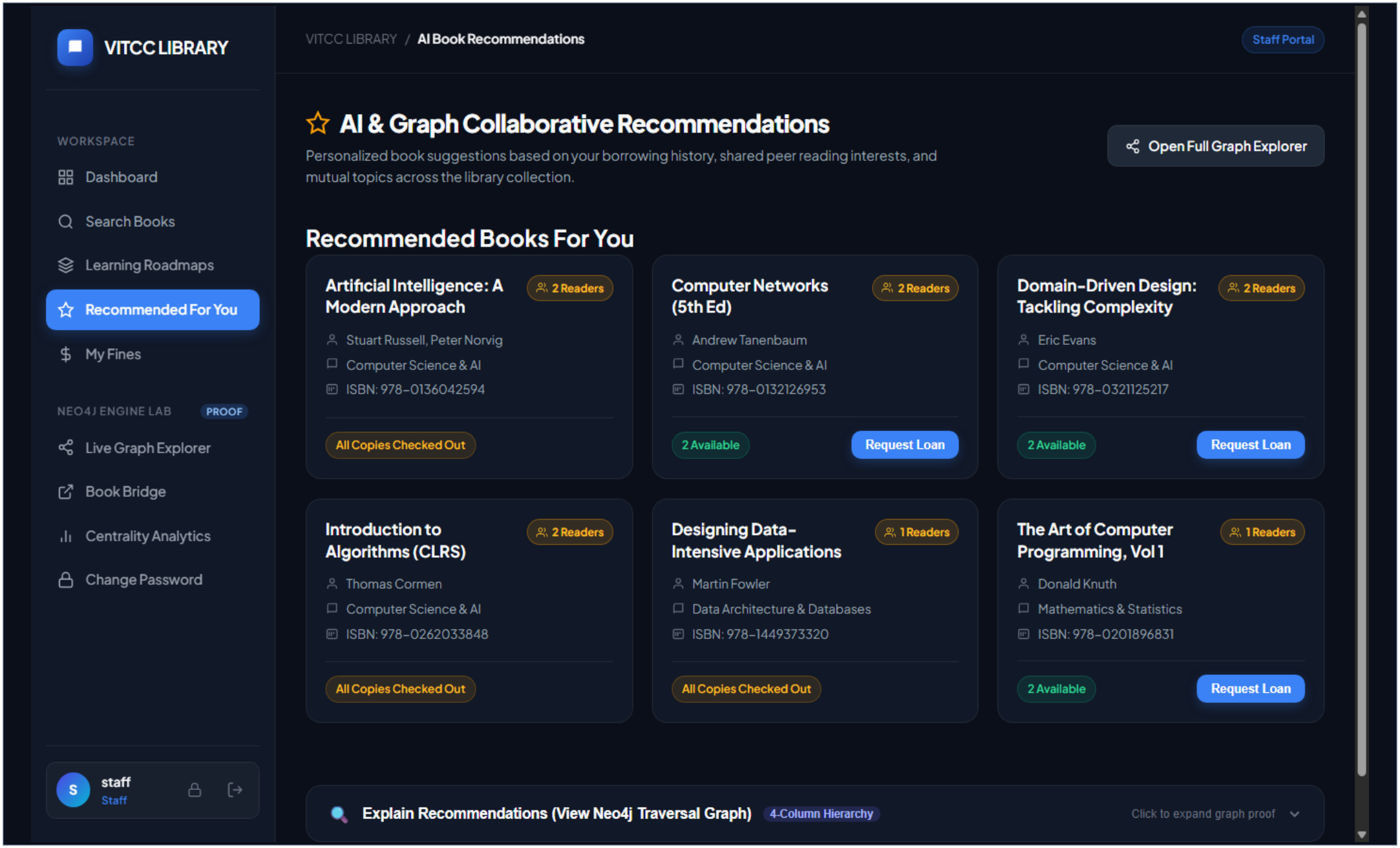
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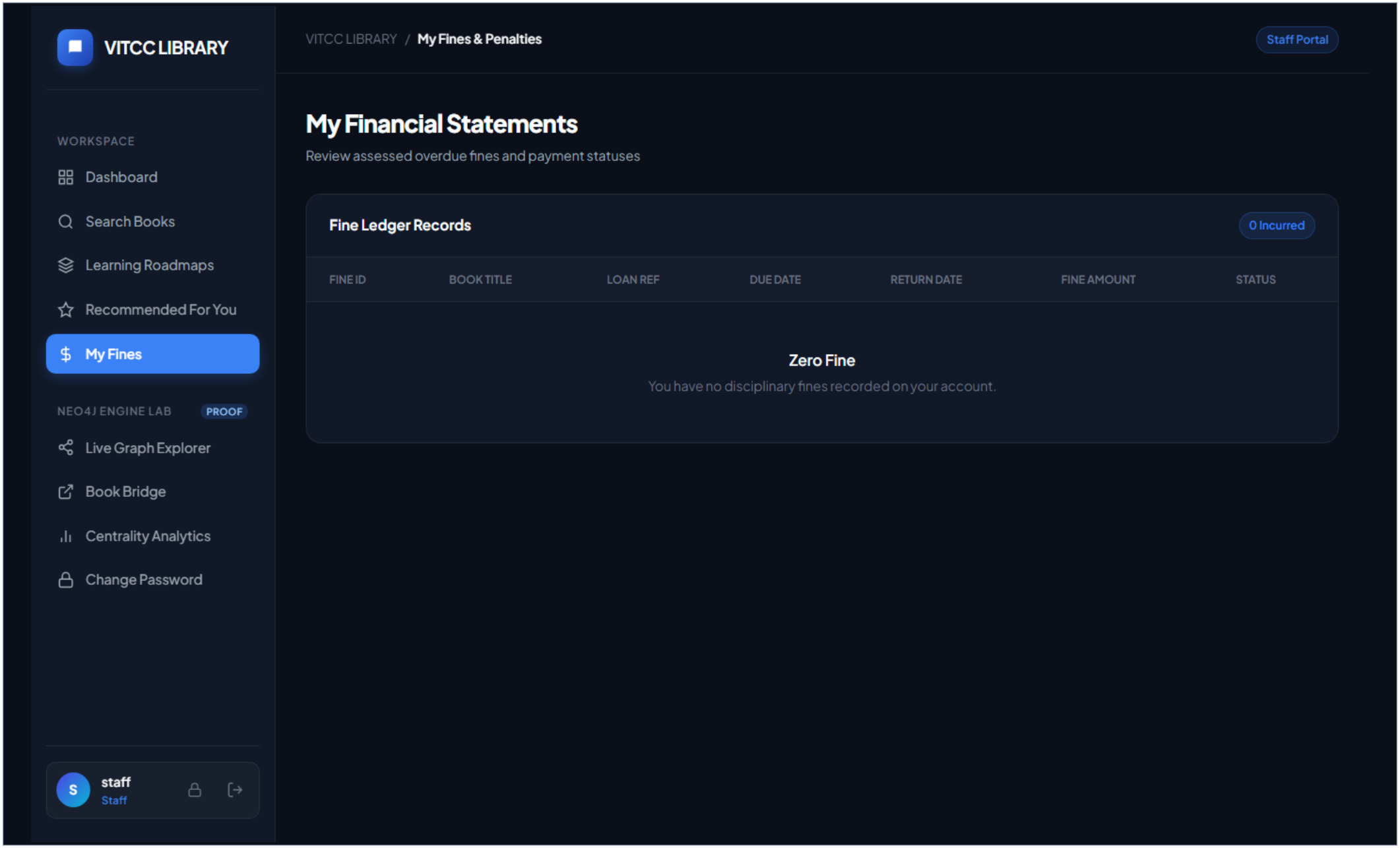

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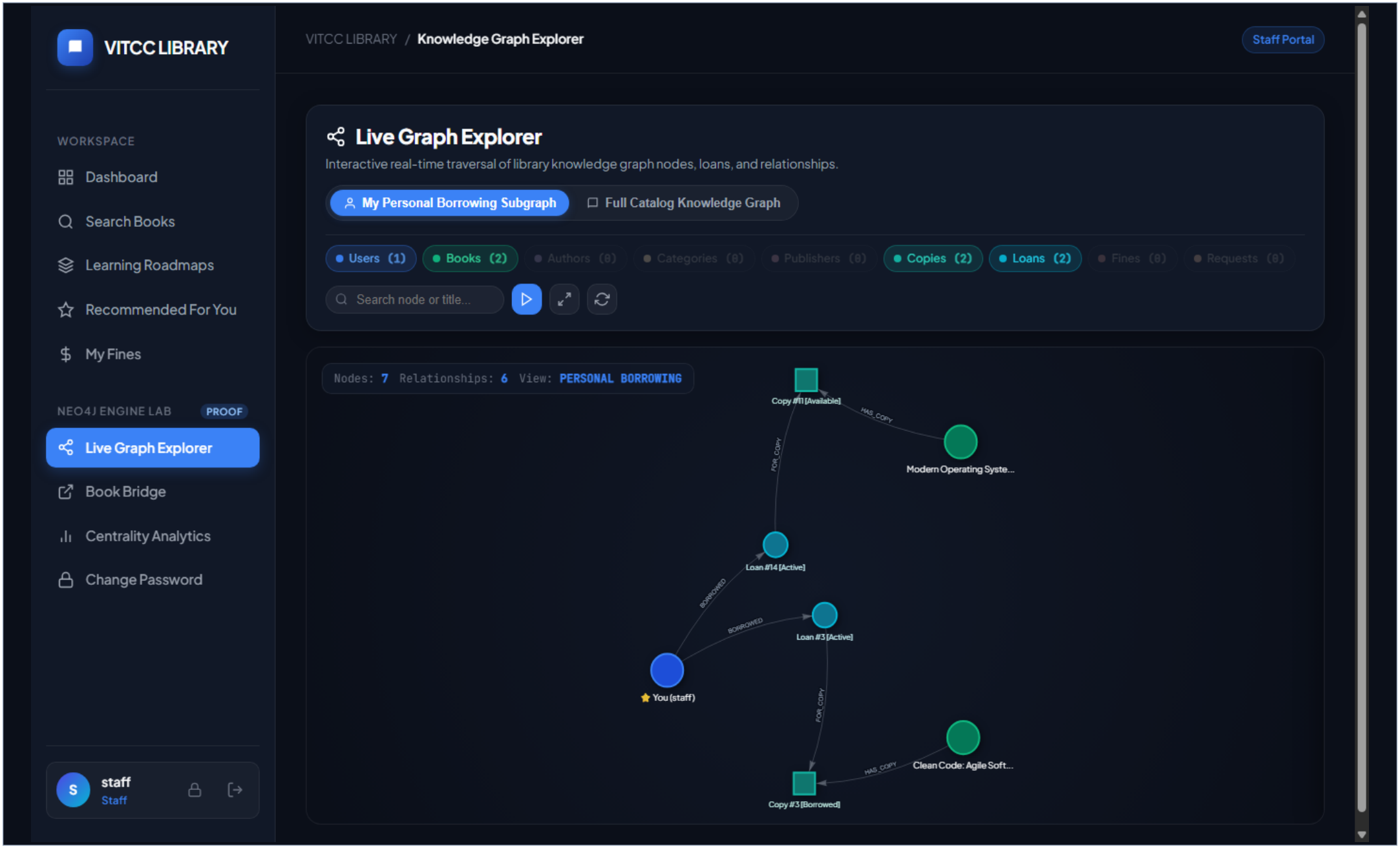
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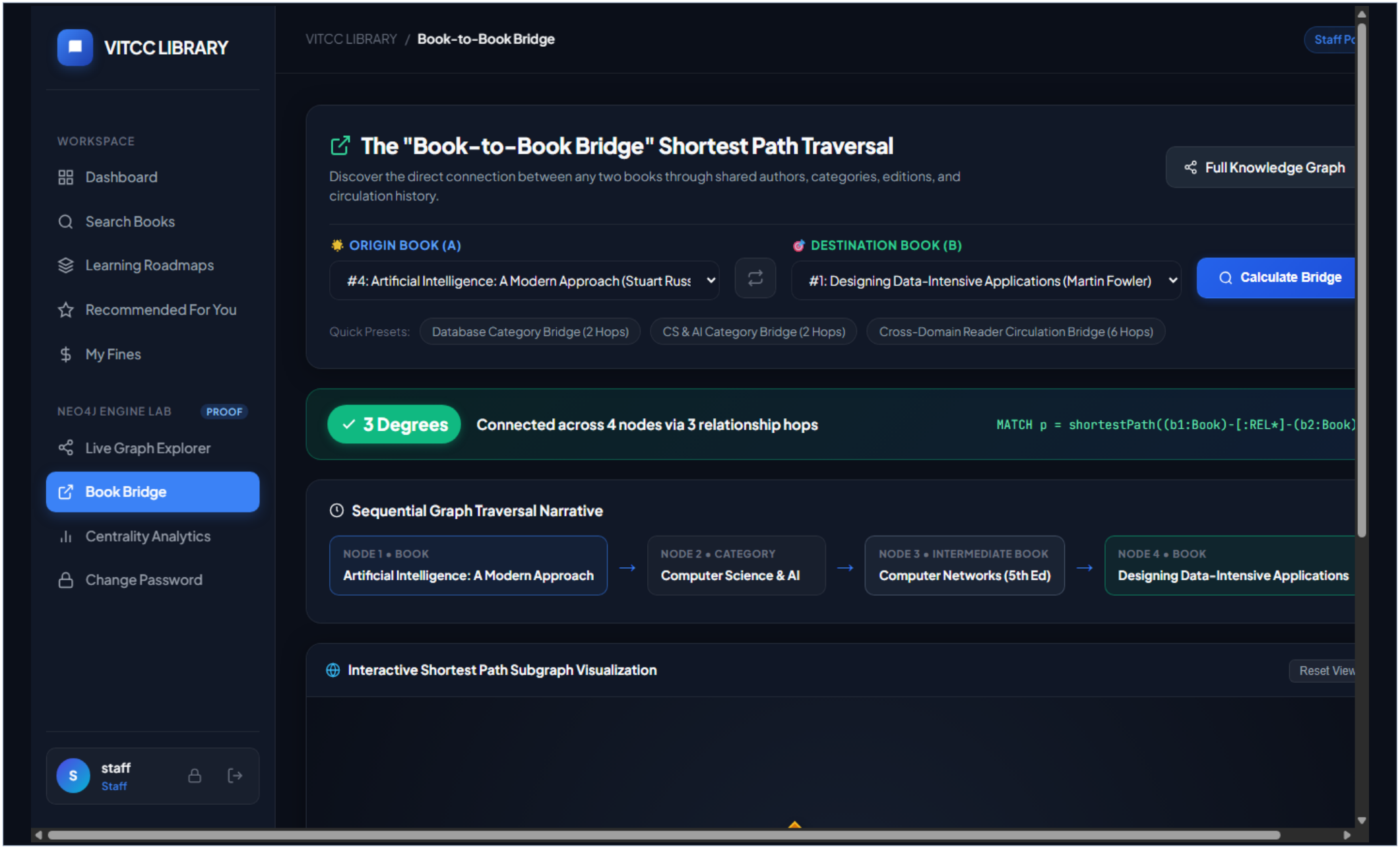
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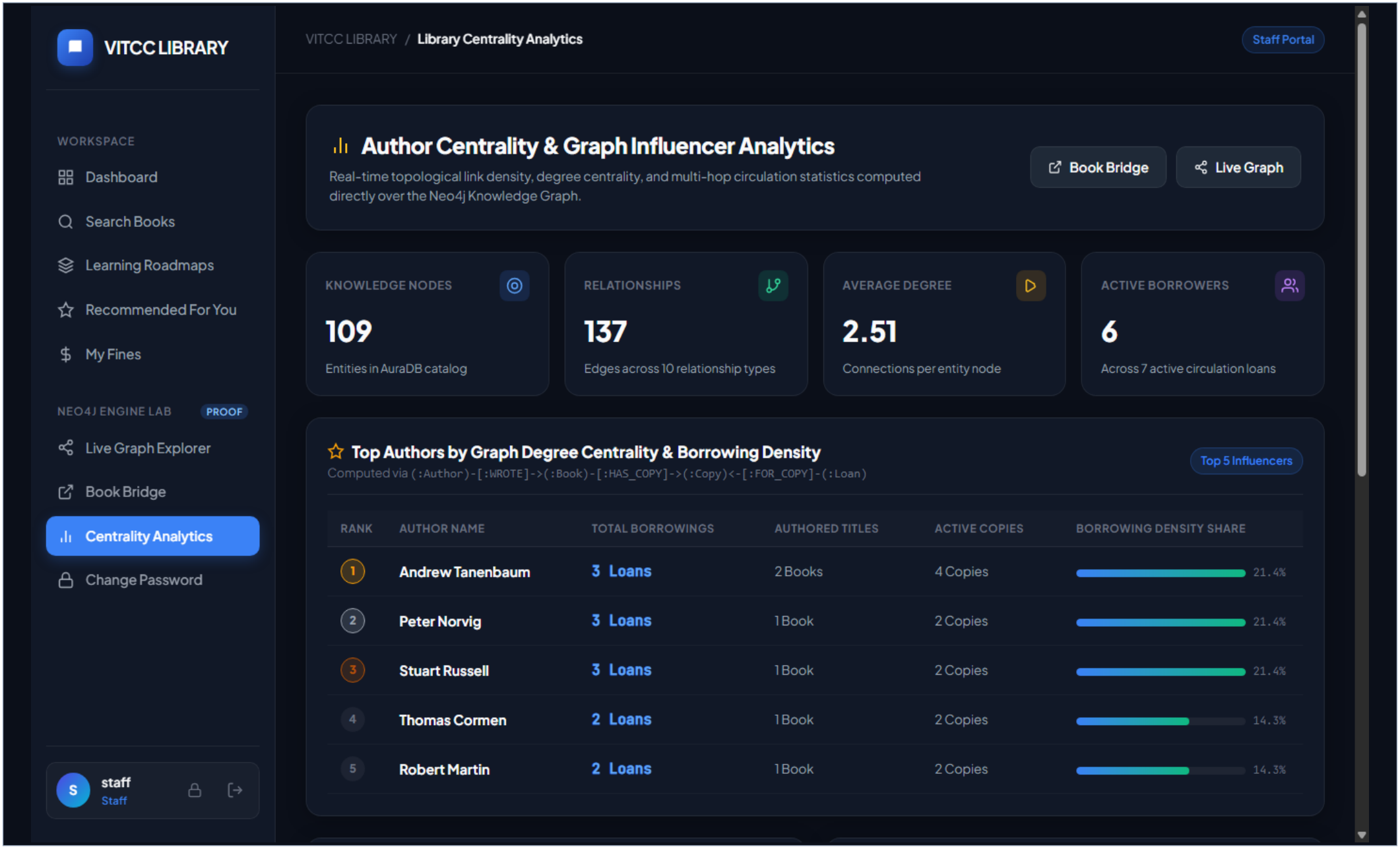
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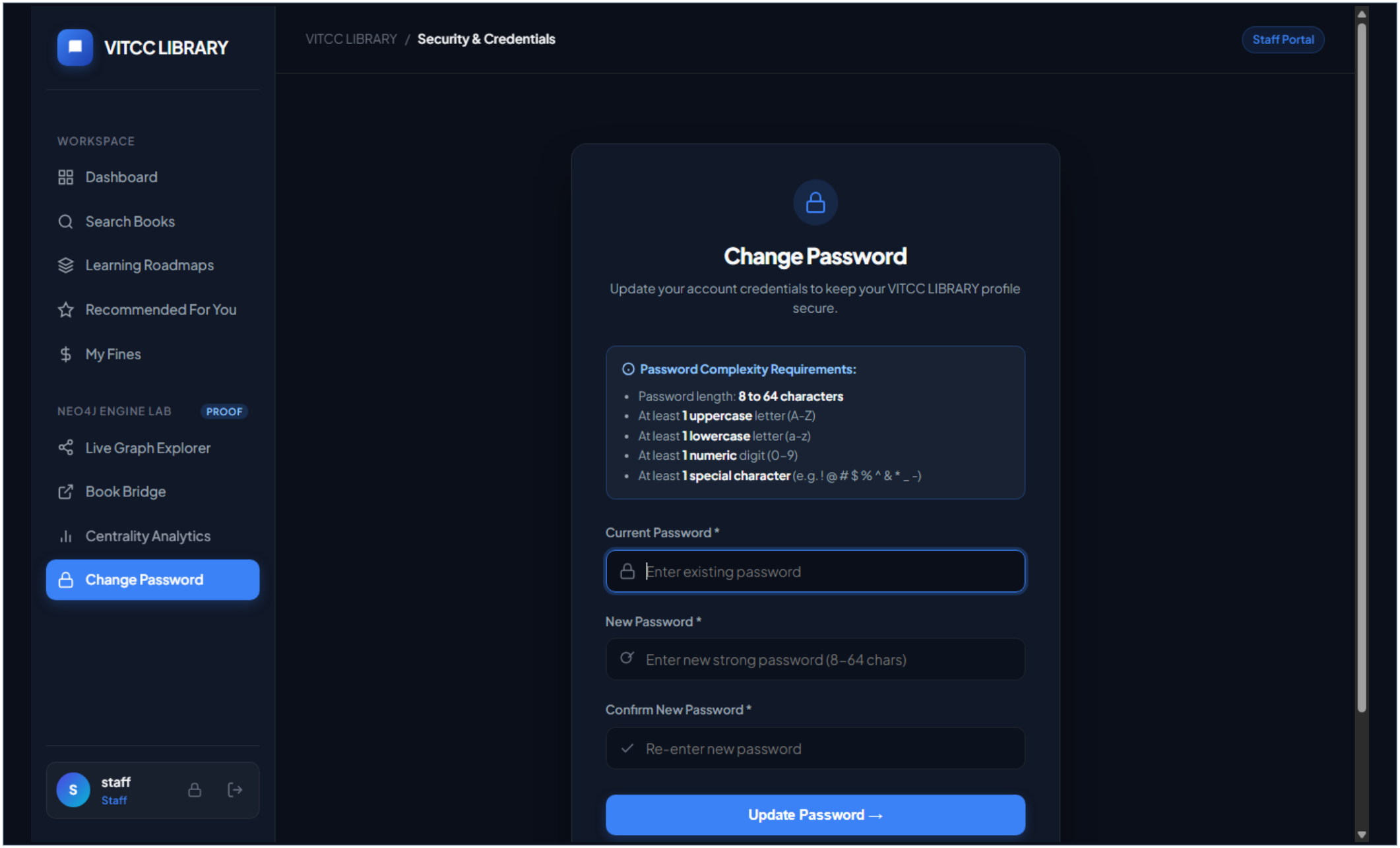
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6
Across 7 active circulation loans
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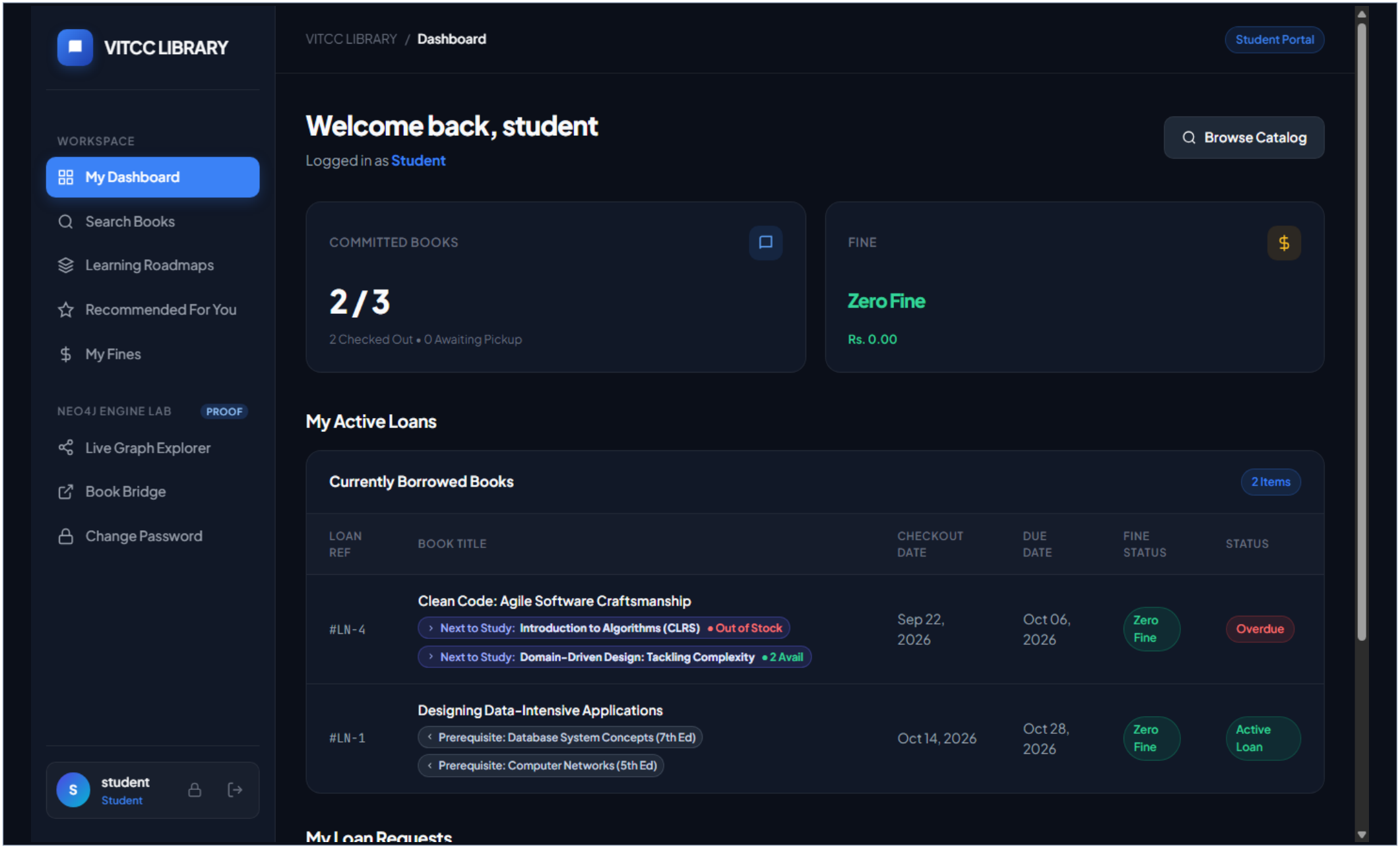

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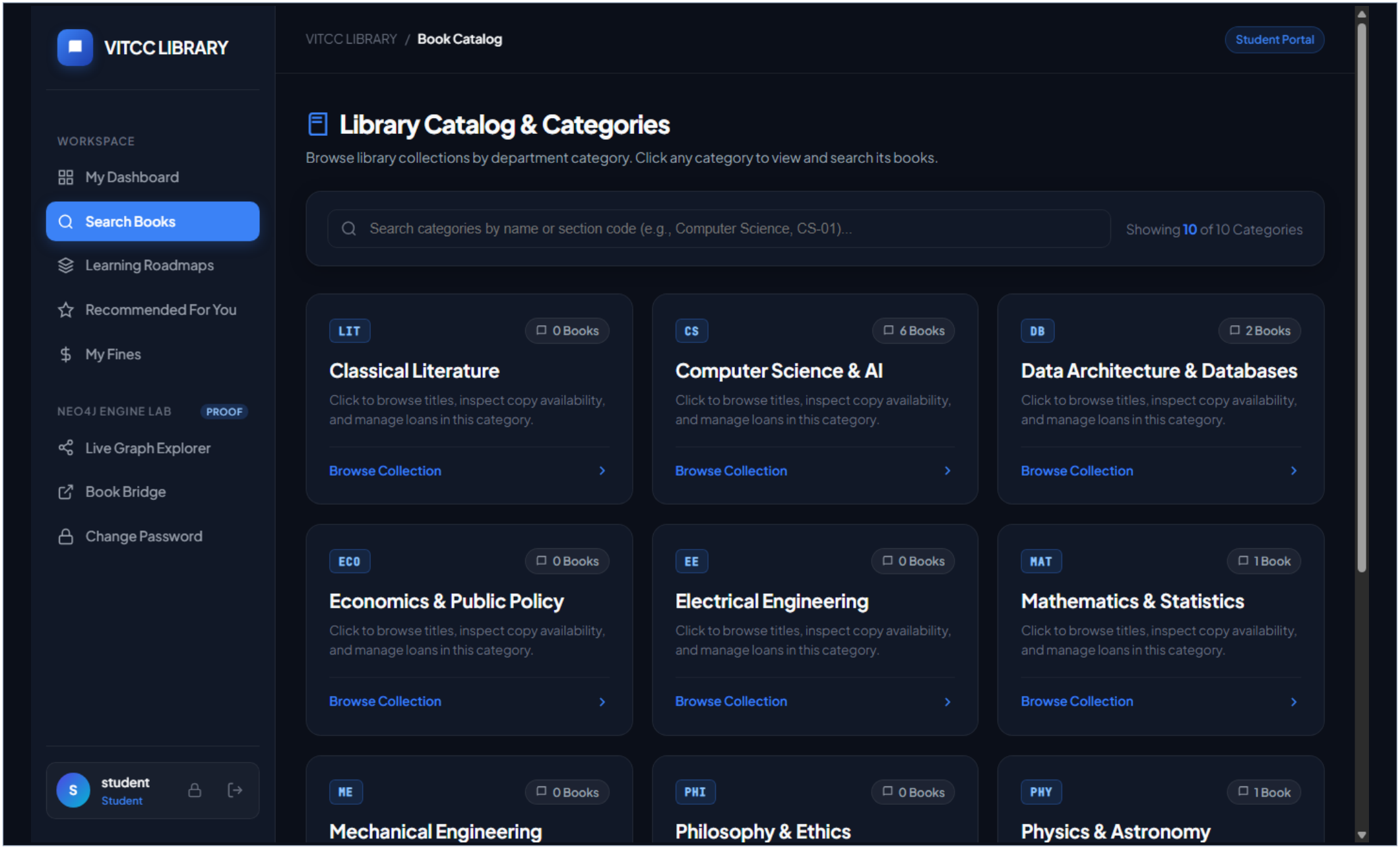
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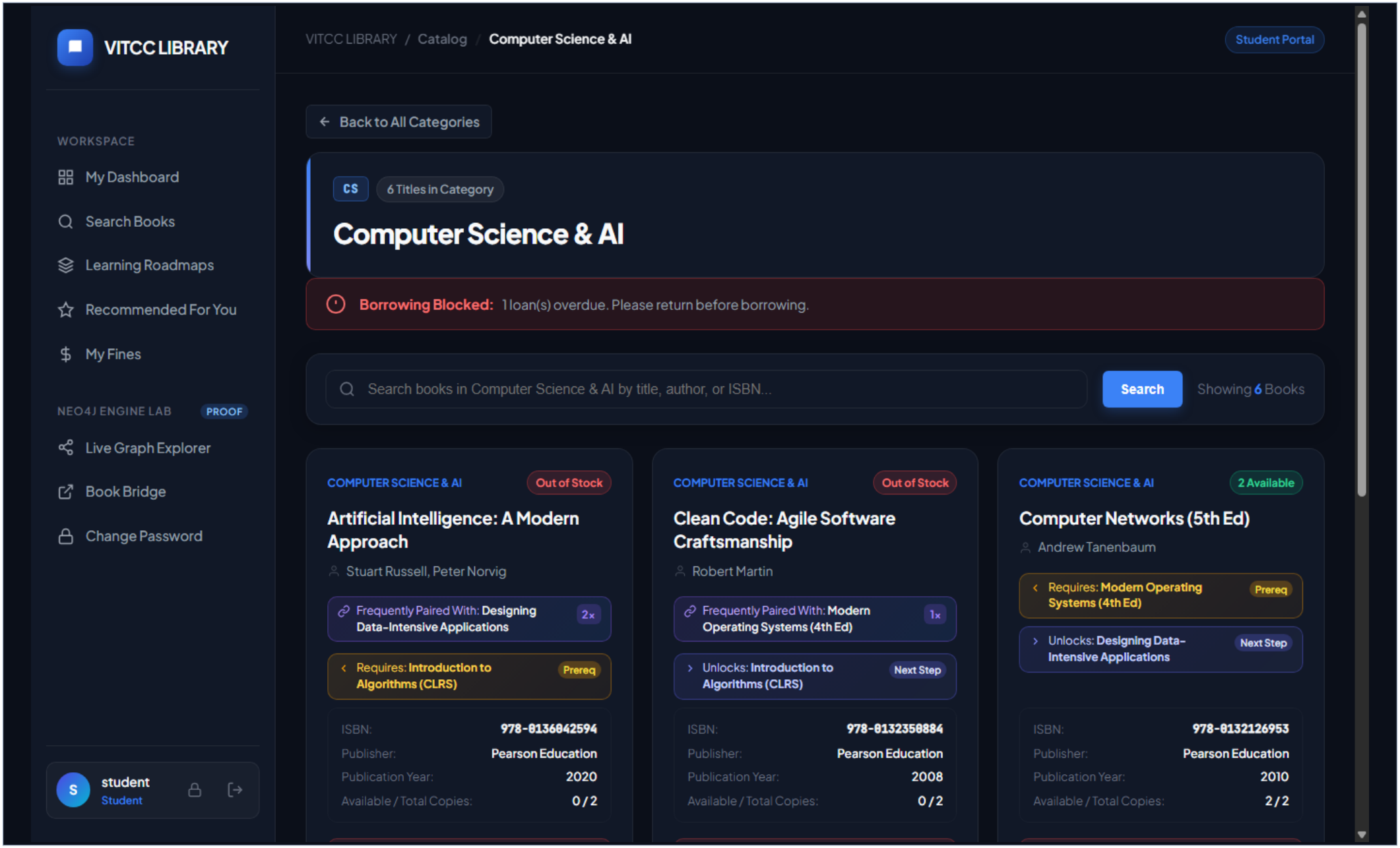
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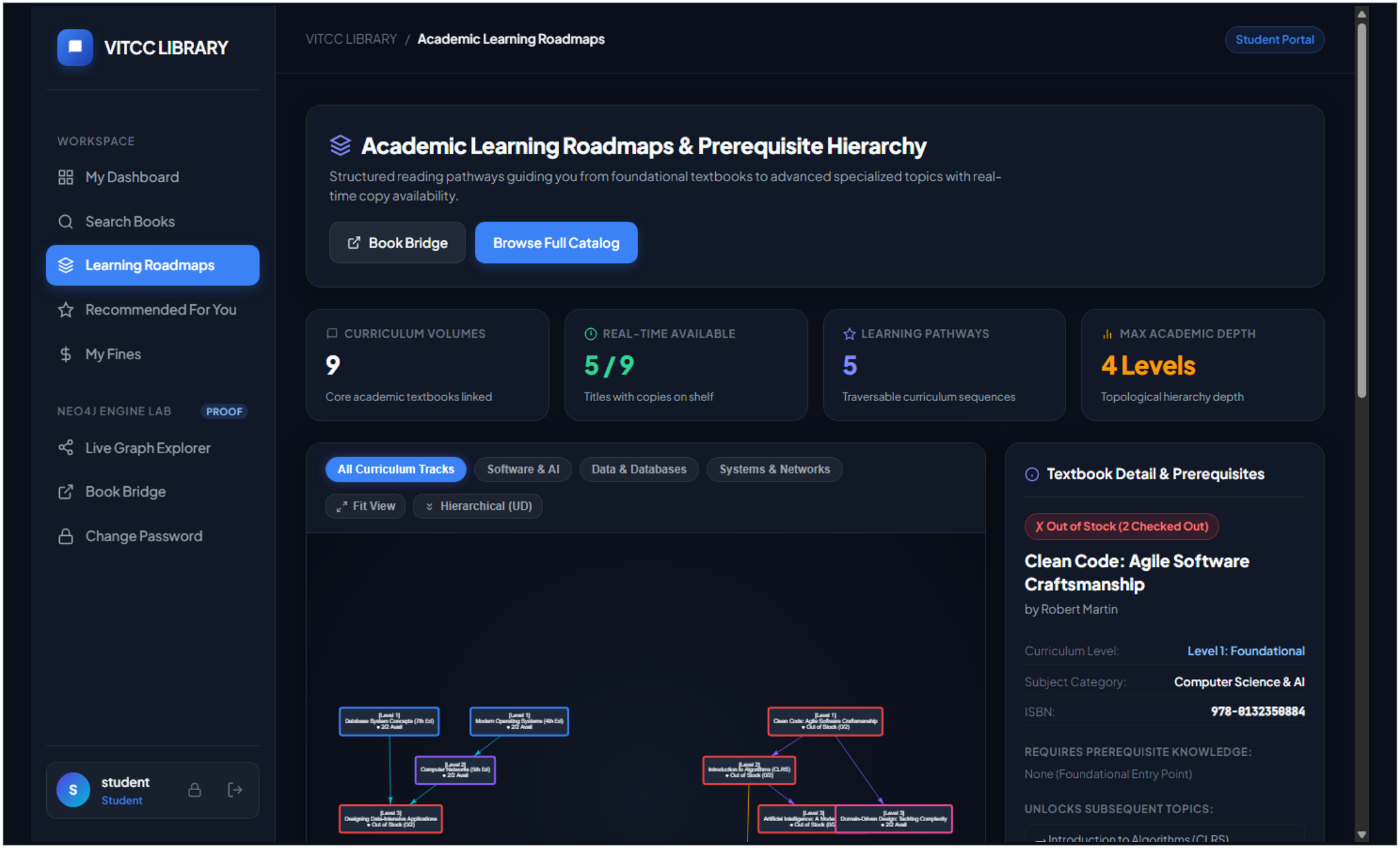
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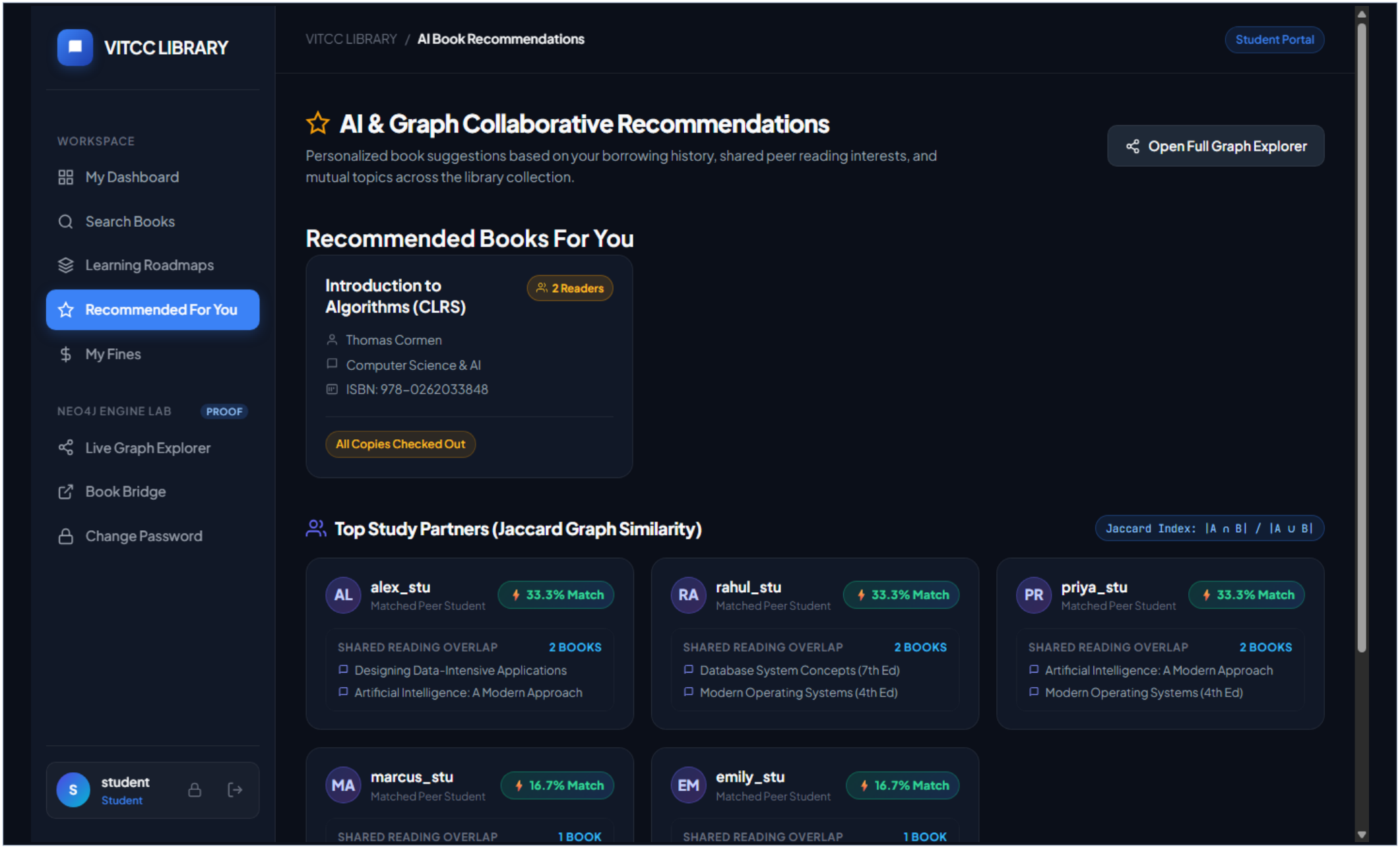
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Subject Category:
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ISBN:
978-0132350884
REQUIRES PREREQUISITE KNOWLEDGE:
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UNLOCKS SUBSEQUENT TOPICS:

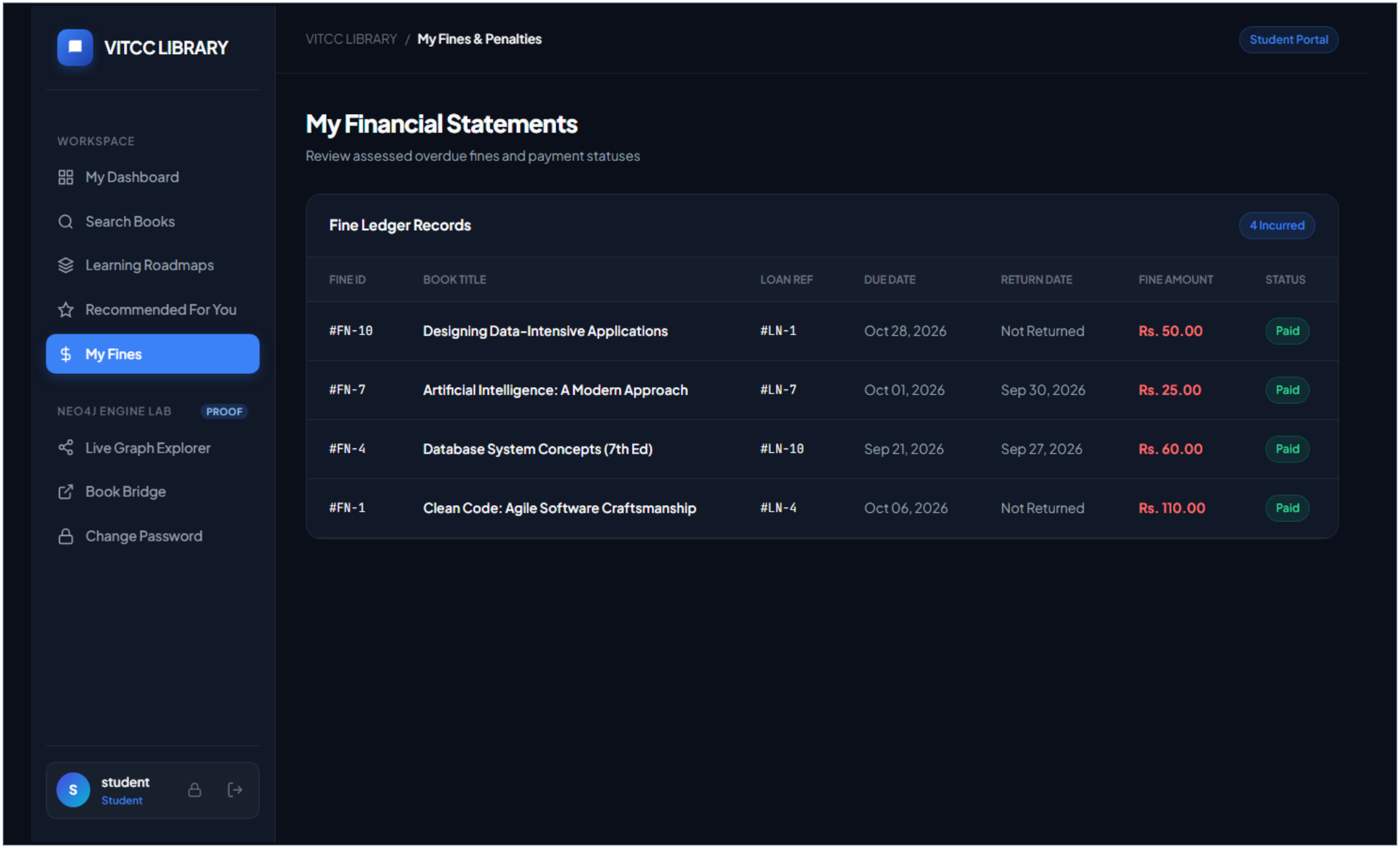

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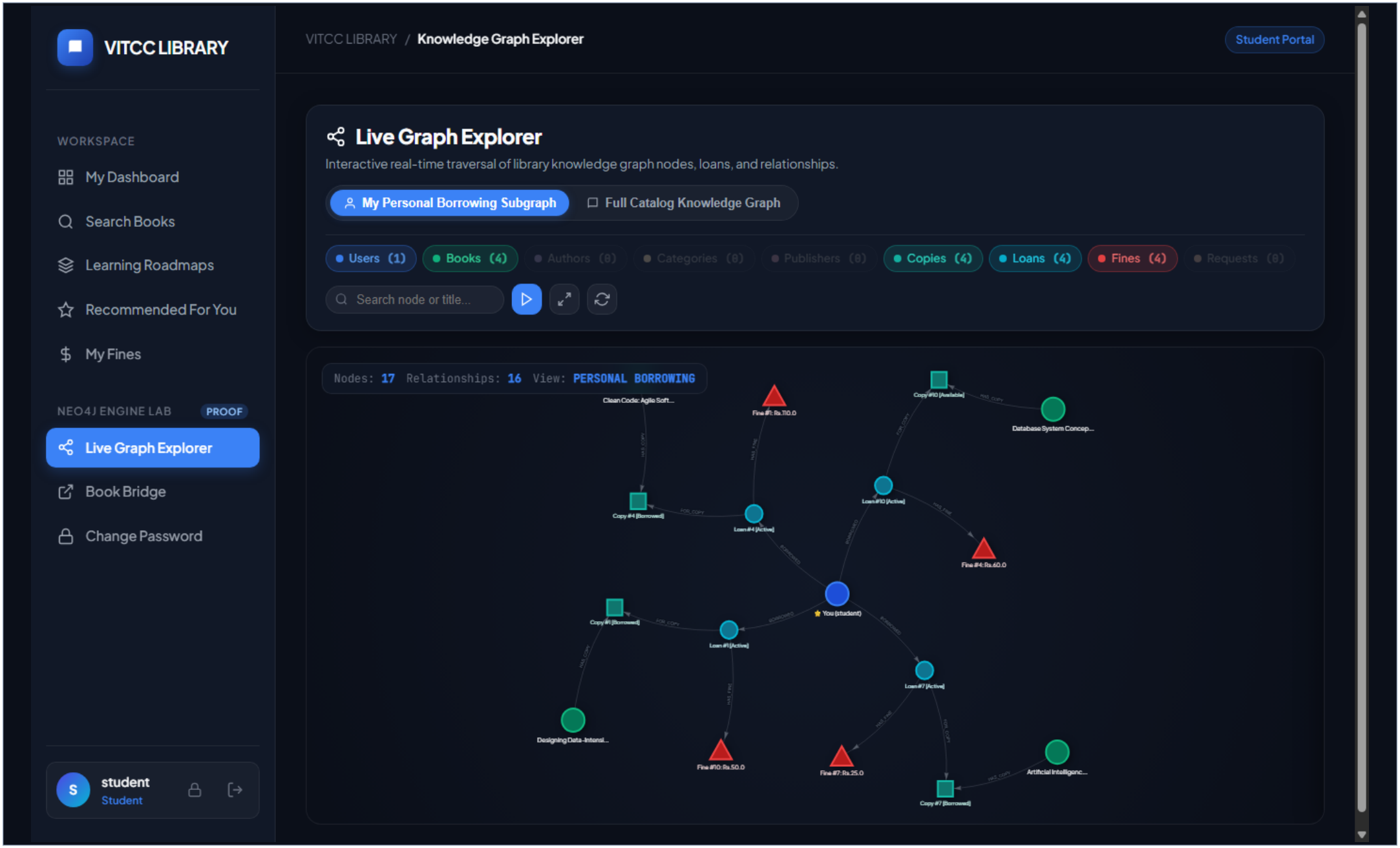
Personal Overdue Fines & Payment History

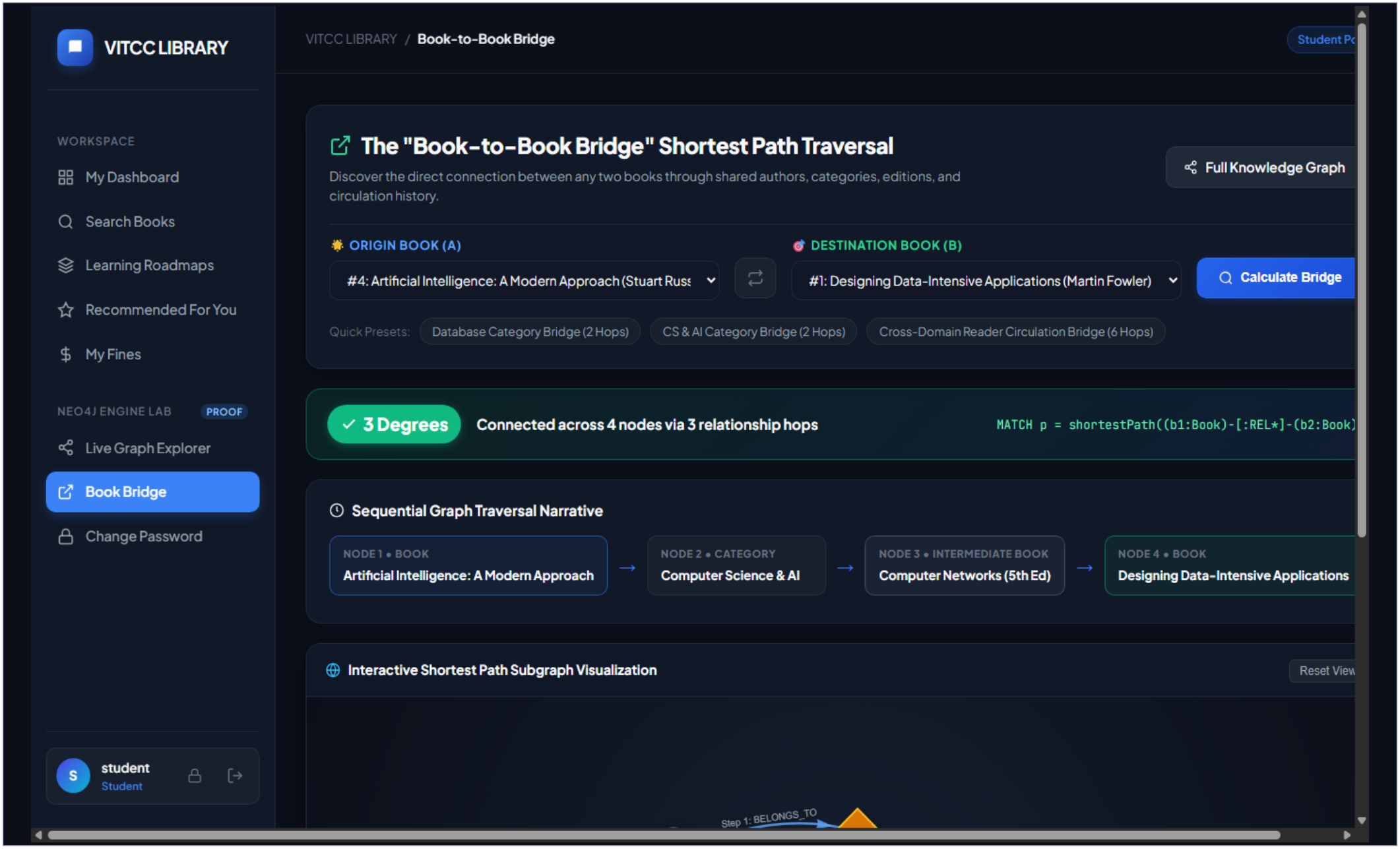
● STUDENT ACCESS

Route: http://127.0.0.1:5000/my_fines | **ID:** 41_student_my_fines

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Books (4)
Authors (0)
Categories (0)
Publishers (0)
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Requests (0)
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CS & AI Category Bridge (2 Hops)
Cross-Domain Reader Circulation Bridge (6 Hops)
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Connected across 4 nodes via 3 relationship hops
MATCH p = shortestPath((b1:Book)-[:REL*]-(b2:Book)
Sequential Graph Traversal Narrative
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NODE 2 • CATEGORY
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NODE 3 • INTERMEDIATE BOOK
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NODE 4 • BOOK
Designing Data-Intensive Applications
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Reset View
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